# DIY Neuroscience Kit Pro

Advanced DIY Neuroscience Kit with BioAmp EXG Pill and Muscle BioAmp Shield

## Overview

This is the advanced DIY Neuroscience Kit with Muscle BioAmp Shield - an EMG Shield for Arduino UNO! The shield offers multiple plug-and-play options, and additional data acquisition while the BioAmp EXG Pill enhances versatility by enabling the measurement of various biopotential signals, including EEG (brain), EOG (eyes), EMG (muscles), and ECG (heart). Start your journey of discovery, as you build exciting projects, develop HCI/BCI applications, and gain valuable insights.

## Kit Contents

- 2 Channel Brain BioAmp Band
- Muscle BioAmp Band
- Heart BioAmp Band
- BioAmp EXG Pill
- BioAmp V3 Cable (100 cm)
- Jumper Cables (x2)
- Arduino Uno R4 with Type-C Cable
- NuPrep Gel (25g)
- Electrode Gel Pouch (100 ml)
- Alcohol Swabs (20 pcs)
- Baby Gel Electrodes (100 pcs)
- Servo claw
- Muscle BioAmp Shield Kit: Muscle BioAmp Shield, Stemma cable, Aux Cable, Muscle BioAmp Band, 24 Pcs Gel Electrodes, BioAmp V3 Cable

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# 1 DIY Neuroscience Kit Pro

## 1.3 Software requirements

To use your DIY Neuroscience Kit Pro, you will need the softwares mentioned below. Instructions on how to use them are provided later in the guide.

- Arduino IDE v1.8.19 (legacy IDE) (Download this to upload Chords arduino firmware to your development board)
- Chords Web (Use this open-source web application to visualize your biopotential signals)
- Visual Studio Code (or any other Code editor of your choice)
- Python (To run Chords-Python script)
- Chords Python (Use this open-source python script designed to record and visualize biopotential signals)

**Note**

1. The Chords Arduino firmware is identical for both Chords Web and Chords Python, so you only need to upload the code once, and you're all set.
2. You can choose either Chords Web or Chords Python to record and visualize your biopotential signals based on your needs. If you're curious, you can try both one at a time.

## 1.4 Using the kit

DIY Neuroscience Kit Pro includes 2 biopotential sensors:

1. BioAmp EXG Pill (Assembled)
2. Muscle BioAmp Shield v0.3 (Assembled)

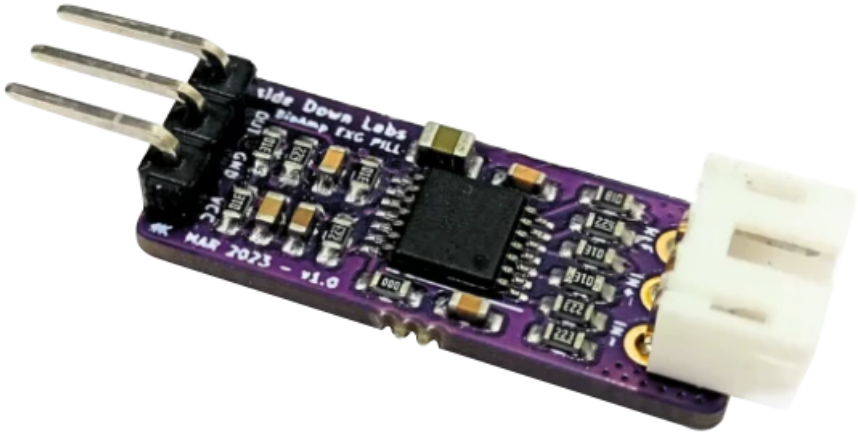
You can use these sensors either separately or connect them together to create a 2-channel EXG acquisition system wherein channel 1 can be used to record EMG signals and channel 2 allows you to record all the biopotential signals from your body (EMG, ECG, EOG, EEG).

### 1.4.1 Step 1: Using BioAmp EXG Pill

BioAmp EXG Pill is a small, powerful analog-front-end (AFE) biopotential signal-acquisition board that can be paired with any microcontroller unit (MCU) or single-board computer (SBC) with an analog-to-digital converter (ADC) such as Arduino UNO & Nano, Adafruit QtPy, STM32 Blue Pill, BeagleBone Black, and Raspberry Pi Pico, to name just a few. It also works with any dedicated ADC, like the Texas Instruments ADS1115 and ADS131M0x, among others.

**Note**

Checkout the complete documentation on *BioAmp EXG Pill* which includes how to use the sensor, record various biopotential signals from your body (ECG, EMG, EOG, EEG) and make different HCI/BCI projects using it.




### 1.4.2 Step 2: Using Muscle BioAmp Shield

Muscle BioAmp Shield is an all-in-one Arduino Uno ElectroMyography (EMG) shield for learning neuroscience with ease which is inspired from BackYard Brains (BYB) Muscle Spiker Shield and provides similar features like hobby servo output, user buttons, LED Bar, Audio output, and battery input. It is perfect for beginners as they can easily stack it on top of Arduino Uno to record, visualize and listen to their muscle signals to make amazing projects in the domain of Human-Computer Interface (HCI).

**Note**

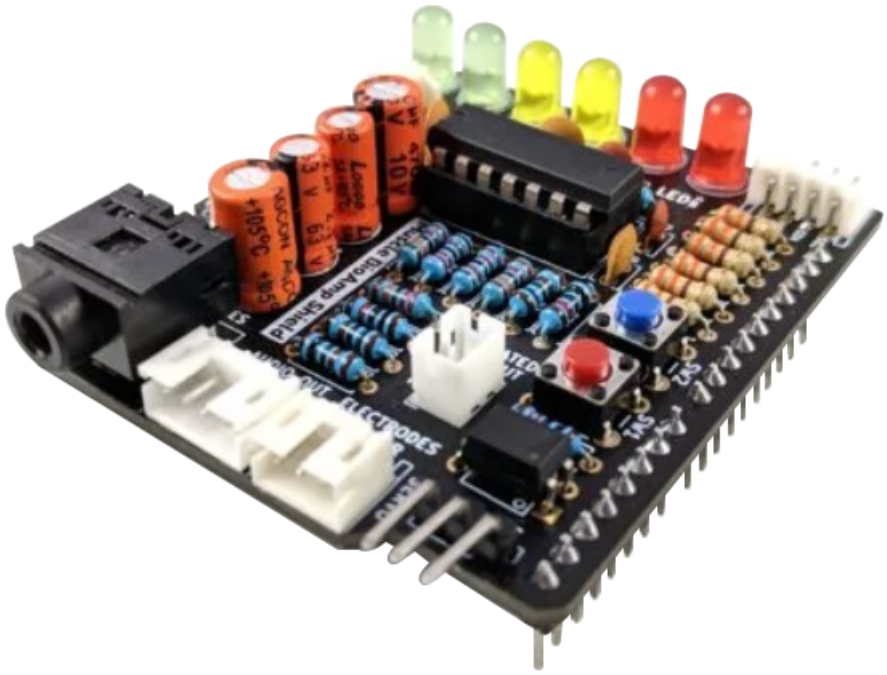
Checkout the complete documentation on *Muscle BioAmp Shield* which includes how to use the sensor, record/visualise/listen muscle signals and make different HCI projects using it.

### 1.4.3 Step 3: Using the sensors together

We believe that you have already gone through the documentation of BioAmp EXG Pill & Muscle BioAmp Shield using the links provided in step 1 and 2 respectively.

You have covered the basic steps till now including connections with the development board, skin preparation, electrodes placements, and recording the signals from your body.

Let's become a PRO and create a 2-channel EXG aquisition system.



### a. Connecting Muscle BioAmp Shield to MCU/ADC

Stack the Muscle BioAmp Shield on top of your Arduino Uno properly.

### b. Configure for ECG/EMG (optional)

BioAmp EXG Pill is by default configured for recording EEG or EOG but if you want to record good quality ECG or EMG, then it is recommended to configure it by making a solder joint as shown in the image.

> **Note**
>
> Even without making the solder joint the BioAmp EXG Pill is capable of recording ECG or EMG but the signals would be more accurate if you configure it.

### c. Connecting sensors together

Connect the BioAmp EXG Pill to the A2 port of Muscle BioAmp Shield via 3-pin STEMMA cable which has JST PH 2.0mm connector on one end and 3 female jumpers on the other end.

| Muscle BioAmp Shield | BioAmp EXG Pill |
|---|---|
| GND | GND |
| VCC | 5V |
| A2 | OUT |

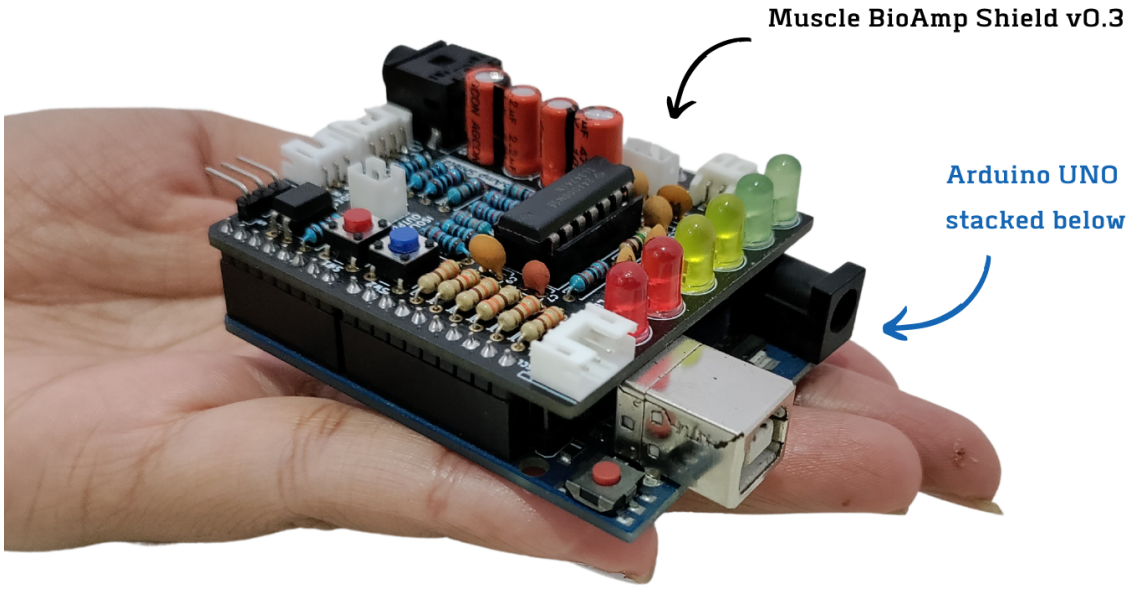



Fig. 1: Stacking Muscle BioAmp Shield on top of Arduino

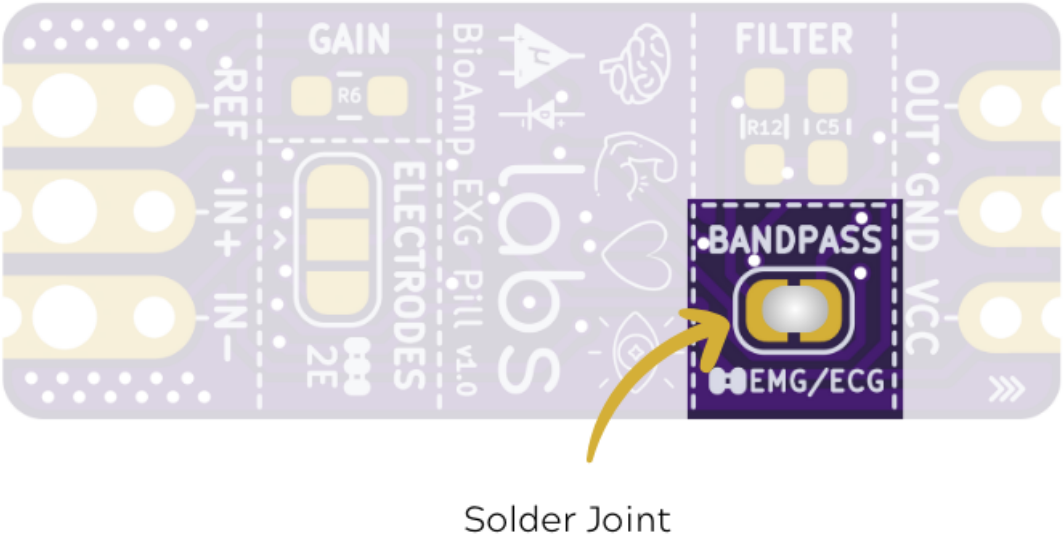



Fig. 2: Configure BioAmp EXG Pill for EMG/ECG

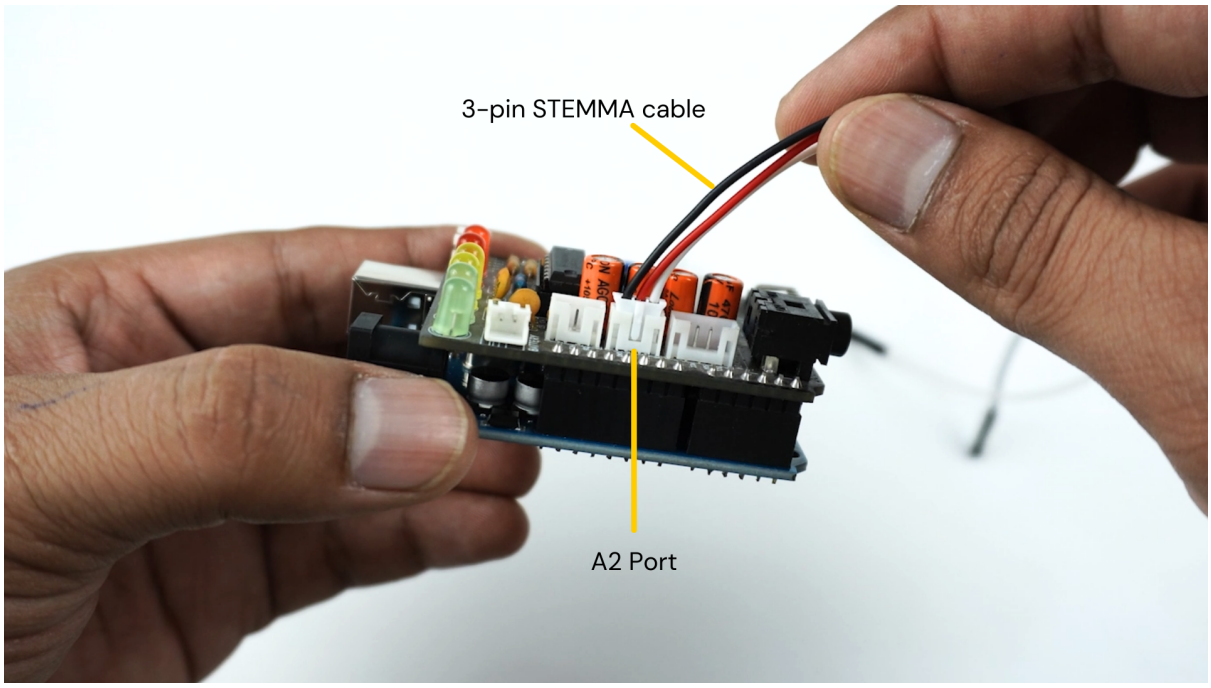



Fig. 3: Inserting JST PH connector in A2 port of Muscle BioAmp Shield

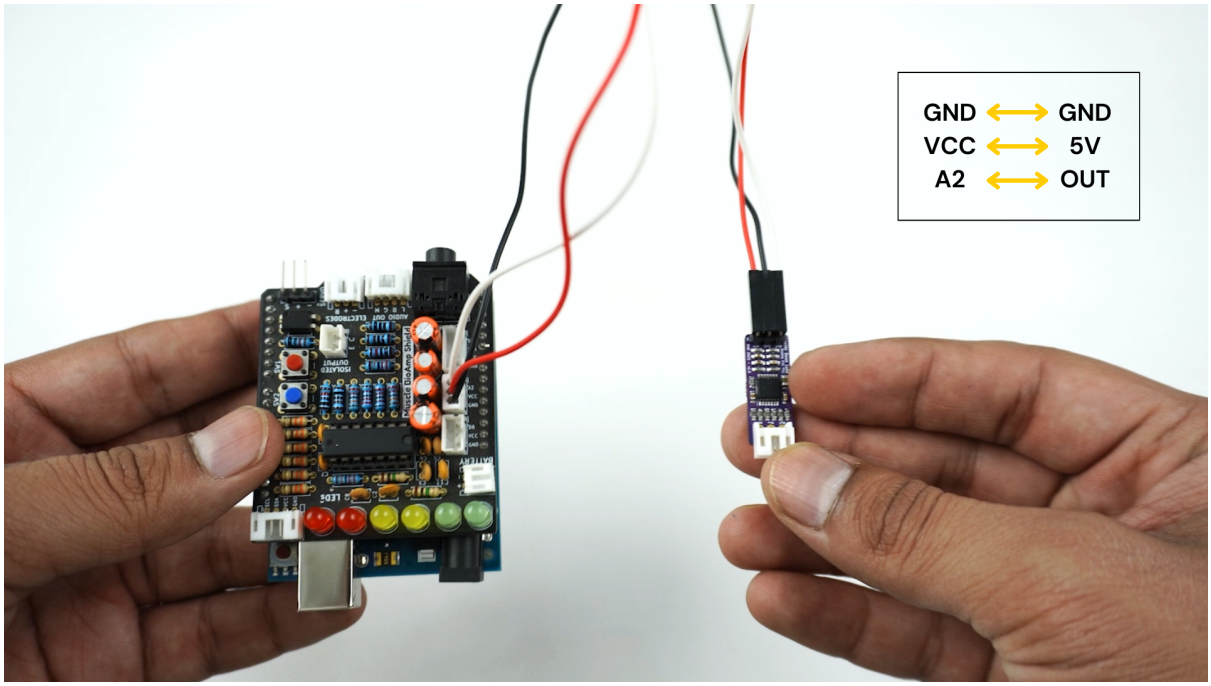



Fig. 4: Muscle BioAmp Shield to BioAmp EXG Pill connections

### d. Connecting electrode cables

Connect one BioAmp cable to BioAmp EXG Pill and another BioAmp cable to Muscle BioAmp Shield by inserting the cable ends into the respective JST PH connectors as shown below:

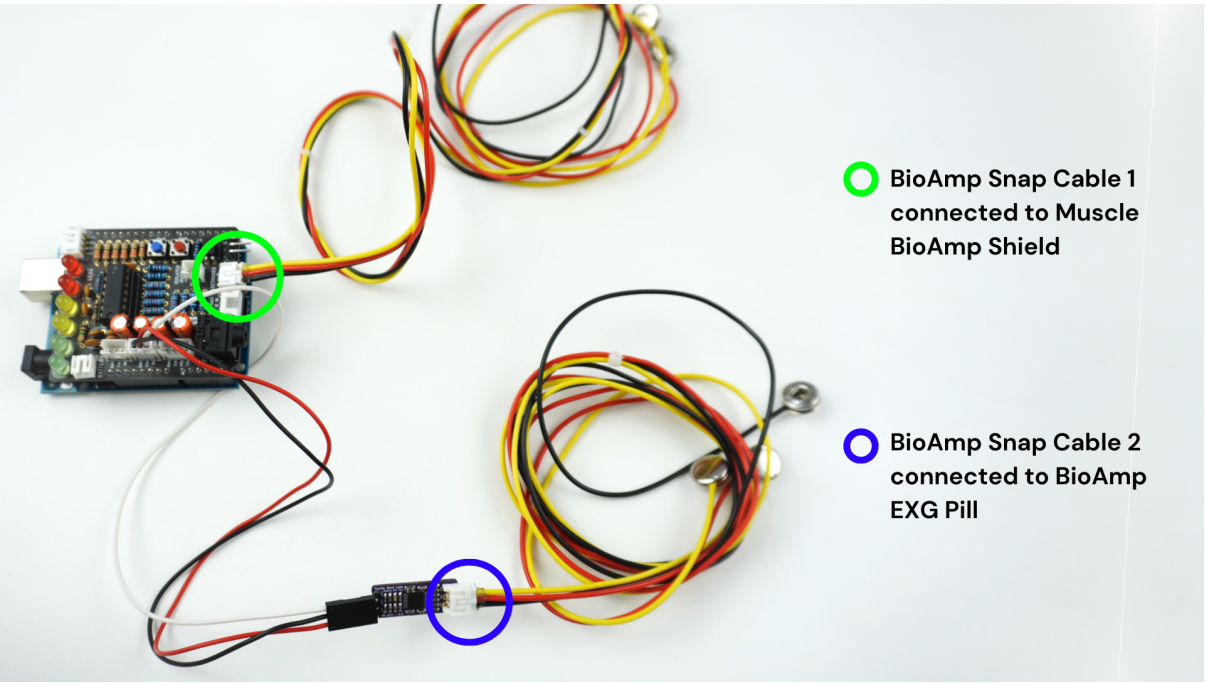



Fig. 5: Connecting BioAmp Cables to the sensors

### e. Skin Preparation

We'll create a 2-channel EMG acquisition system and to do so, we'll be using both the sensors to record EMG signals from the ulnar nerve of both the hands, Thus, prepare the skin accordingly.

Apply Nuprep Skin Preparation Gel on the skin surface where electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step Skin Preparation Guide.

### f. Electrodes placement

We have 2 options to measure the EMG signals, either using the gel electrodes or using dry electrode based Muscle BioAmp Band. You can try both of them one by one.

#### Using gel electrodes

1. Snap the BioAmp Cable connected to BioAmp EXG Pill to gel electrodes.
2. Peel the plastic backing from electrodes.
3. Place the IN+ and IN- cables on the left arm near the ulnar nerve & REF (reference) at the back of your left hand as shown below.
4. Now snap the BioAmp Cable connected to Muscle BioAmp Shield to gel electrodes.
5. Peel the plastic backing from electrodes.

6. Place the IN+ and IN- cables on the right arm near the ulnar nerve & REF (reference) at the back of your right hand as shown below.

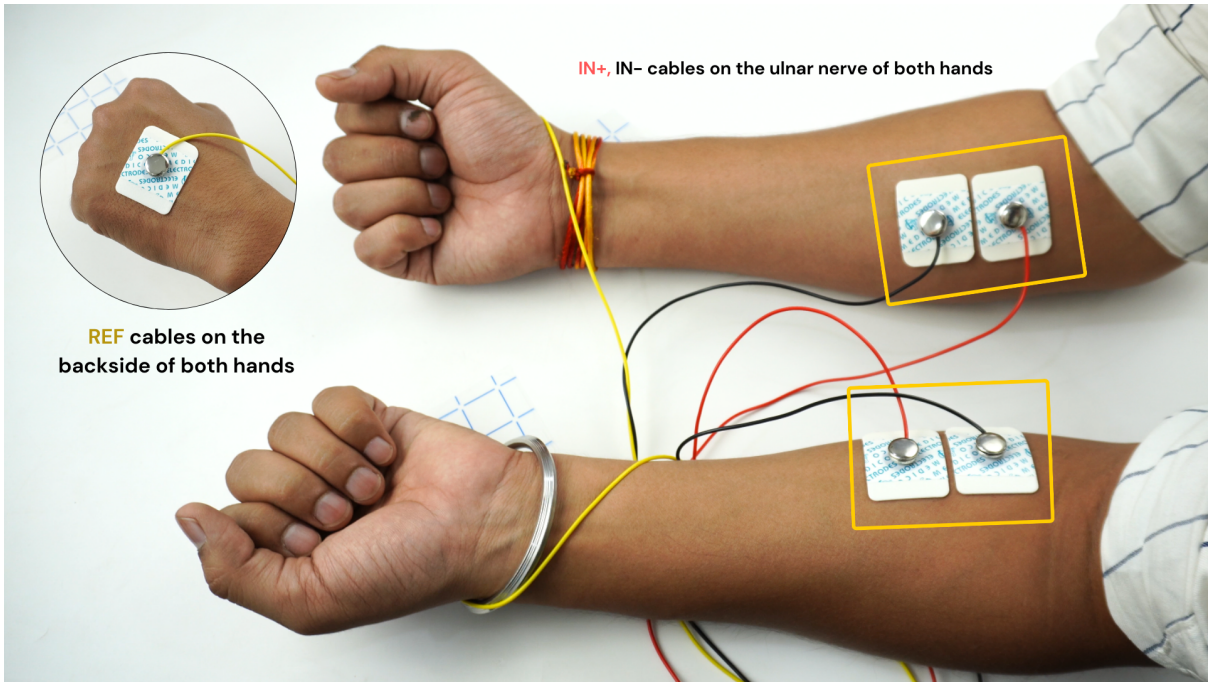



Fig. 6: Gel electrodes placement

### Using Muscle BioAmp Band

1. Snap the BioAmp Cable connected to BioAmp EXG Pill on Muscle BioAmp Band in a way such that IN+ and IN- are placed on the left arm near the ulnar nerve & REF (reference) on the far side of the band.
2. Snap the BioAmp Cable connected to Muscle BioAmp Shield on Muscle BioAmp Band in a way such that IN+ and IN- are placed on the right arm near the ulnar nerve & REF (reference) on the far side of the band.
3. Now put a small drop of electrode gel between the skin and metallic parts of BioAmp Cables to get the best results.

**Tip**

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

**Tutorial on how to use the band:**

https://youtu.be/xYZdw0aesa0

**Note**

In this demonstration we are recording EMG signals from the ulnar nerve, but you can record EMG from other areas as well (biceps, triceps, legs, jaw etc) as per your project requirements. Just make sure to place the IN+, IN- electrodes on the targeted muscle and REF on a bony part.

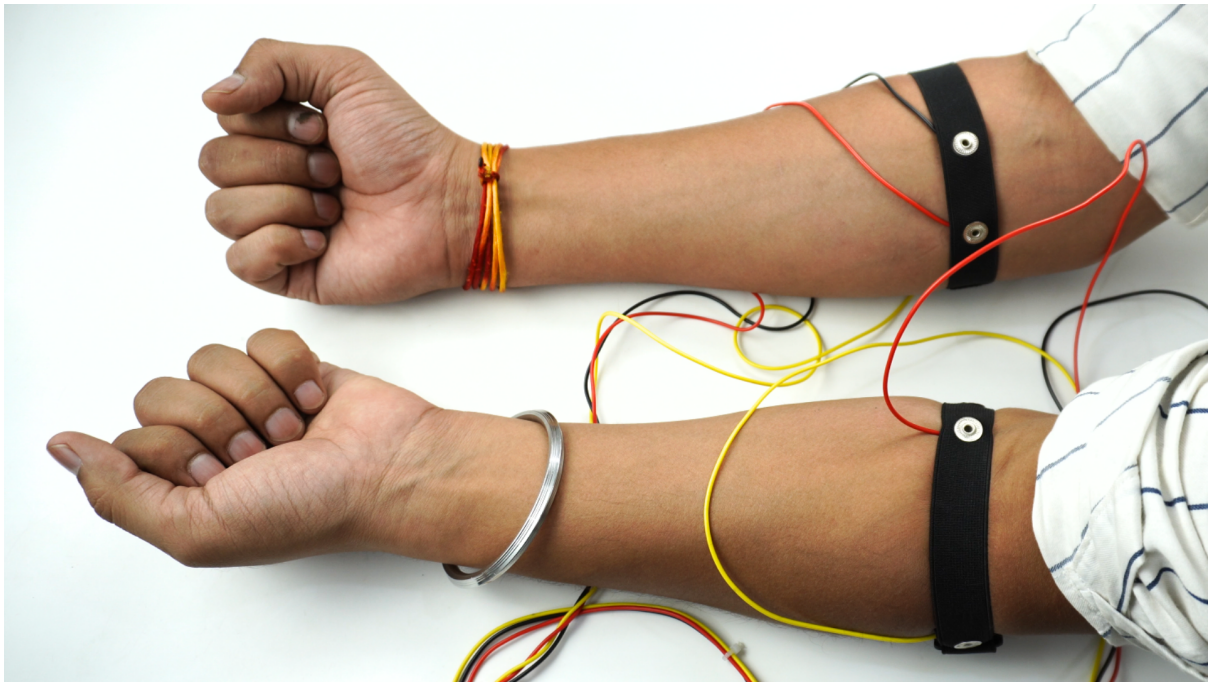


Fig. 7: Muscle BioAmp Band placement

### g. Uploading the code

Connect Arduino Uno to your laptop using the USB cable (Type A to Type B). Download the github repository given below:

Muscle BioAmp Arduino Firmware

Go to the folder `8_EMGScrolling`, open the arduino sketch `8_EMGScrolling.ino` in Arduino IDE.

Go to `tools` from the menu bar, select `board` option then select Arduino UNO. In the same menu, select the COM port on which your Arduino Uno is connected. To find out the right COM port, disconnect your Arduino UNO board and reopen the menu. The entry that disappears should be the right COM port. Now upload the code.

**Important**

Make sure your laptop is not connected to a charger and sit 5m away from any AC appliances for best signal acquisition.

### h. Testing the connections

Go to `tools` from the menu bar, click on `serial monitor` to open it or click on the icon on the top right corner. Try flexing both of your arms one-by-one. The output values should be 0 at this point.

Press the `SW1 button` on Muscle BioAmp Shield. Now you'll see green LED glowing on the LED bar. When you flex your right arm, you'll get output value 1 on the serial monitor and red LED will glow. Similarly, when you flex your left arm, you'll get output value 2 and yellow LED will glow.

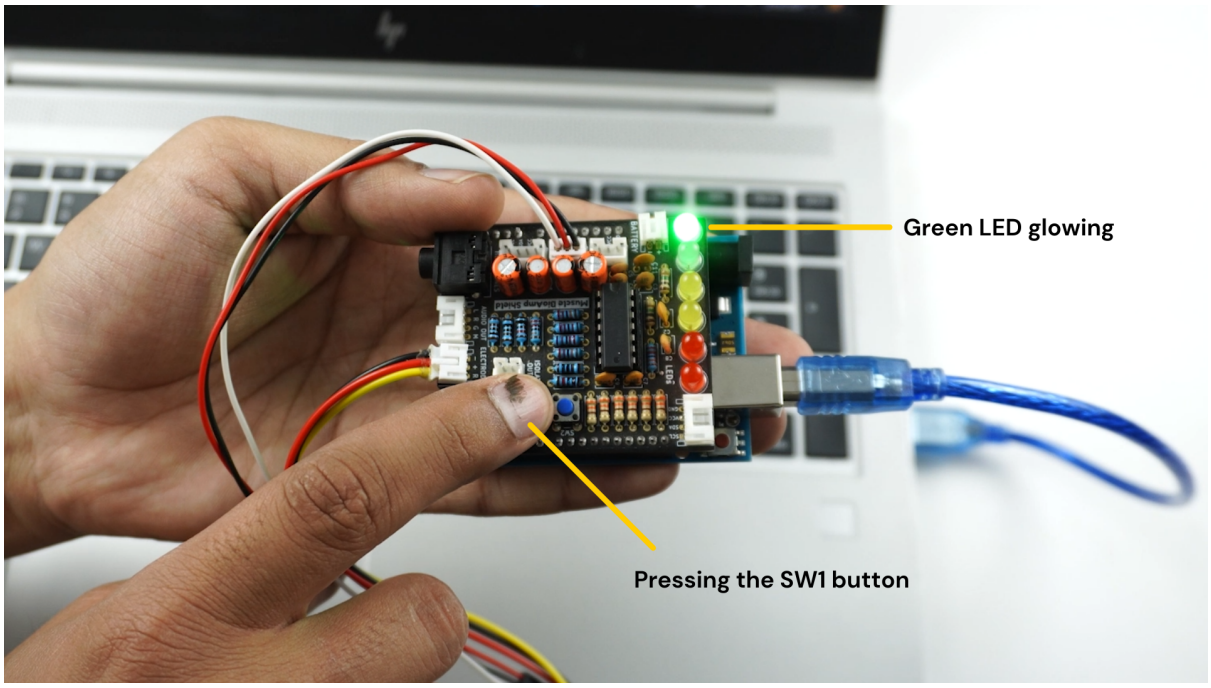



Fig. 8: Press the SW1 button to start getting the output

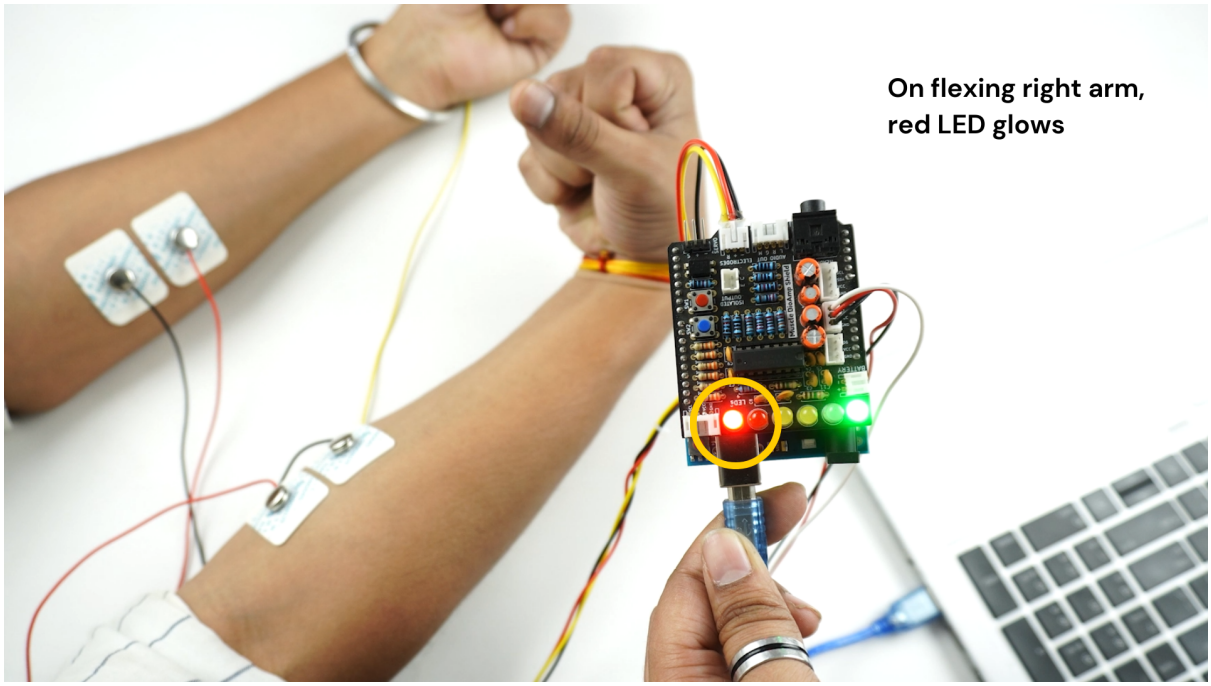



Fig. 9: Flex the right arm, red LED glows

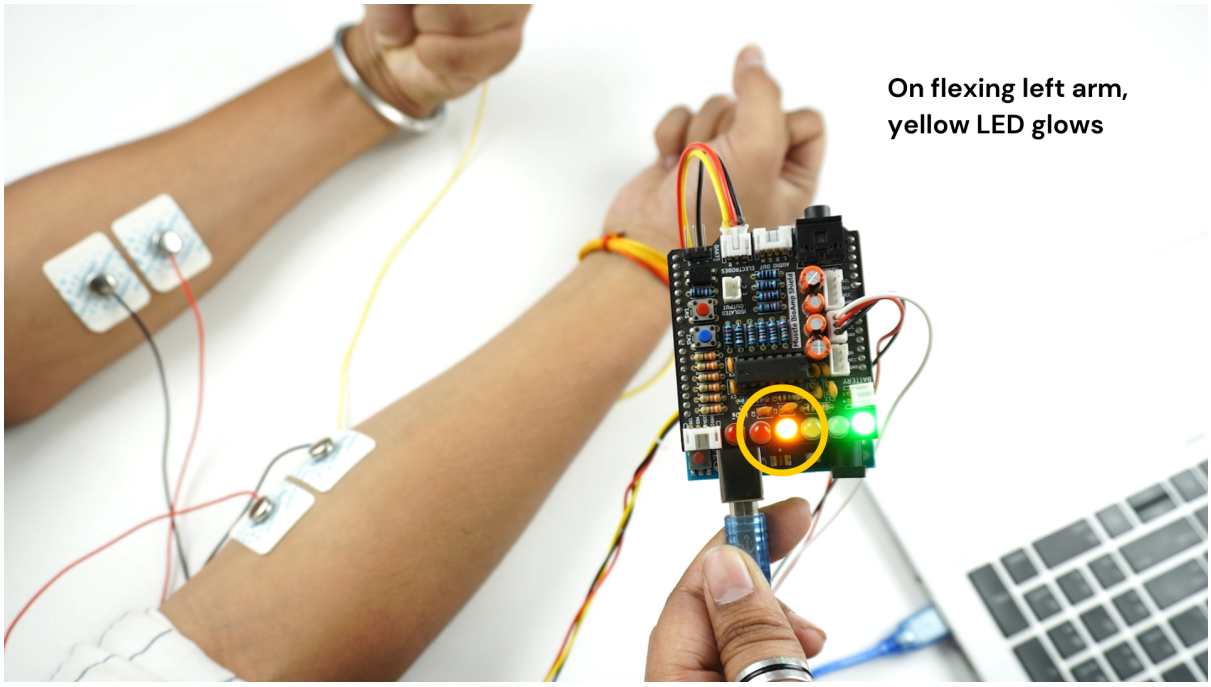



Fig. 10: Flex the left arm, yellow LED glows

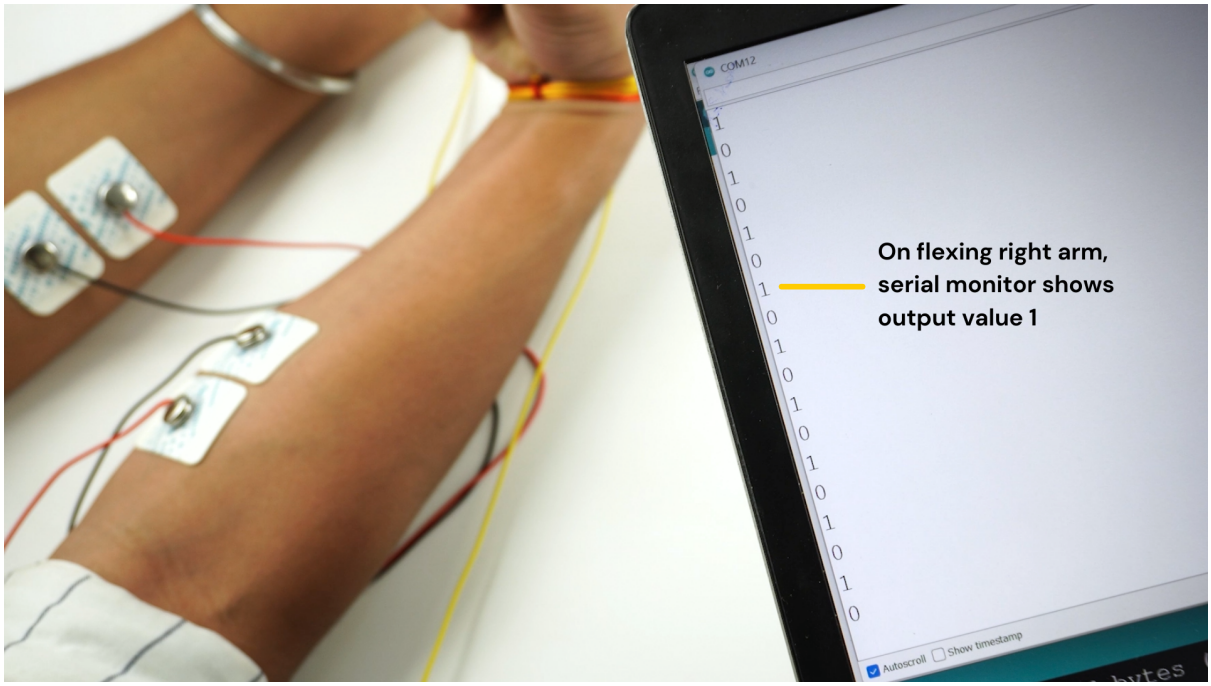



Fig. 11: Flex the right arm, serial monitor shows output value 1

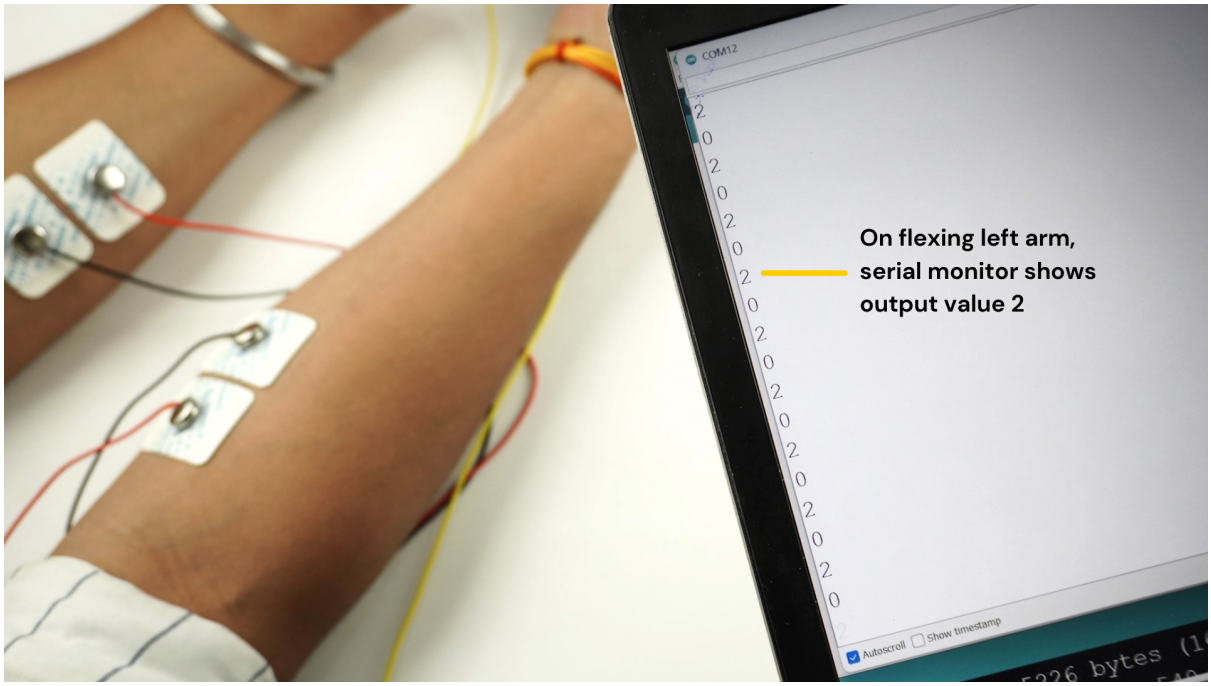



Fig. 12: Flex the left arm, serial monitor shows output value 2

### i. Running python script

Open Visual Studio Code, click on File > Open folder to open the folder `8_EMGScrolling`.

Open the terminal, and ensure that the path is configured to the location of the `requirement.txt` file.

To install all the modules that are required to run the Python script, write the given command in the terminal:

```
pip install -r requirements.txt
```

Open `EMG_Scroll.py` and change the COM Port in the code (line 14) as per the COM Port you selected in Arduino IDE. Save the file by clicking CTRL + S.

```
# Arduino serial port interface
ser = serial.Serial('COM12', 115200, timeout=1)
```

Run the Python script `EMG_Scroll.py` by writing the given command in the terminal:

```
python EMG_Scroll.py
```

### j. Scrolling using EMG signals

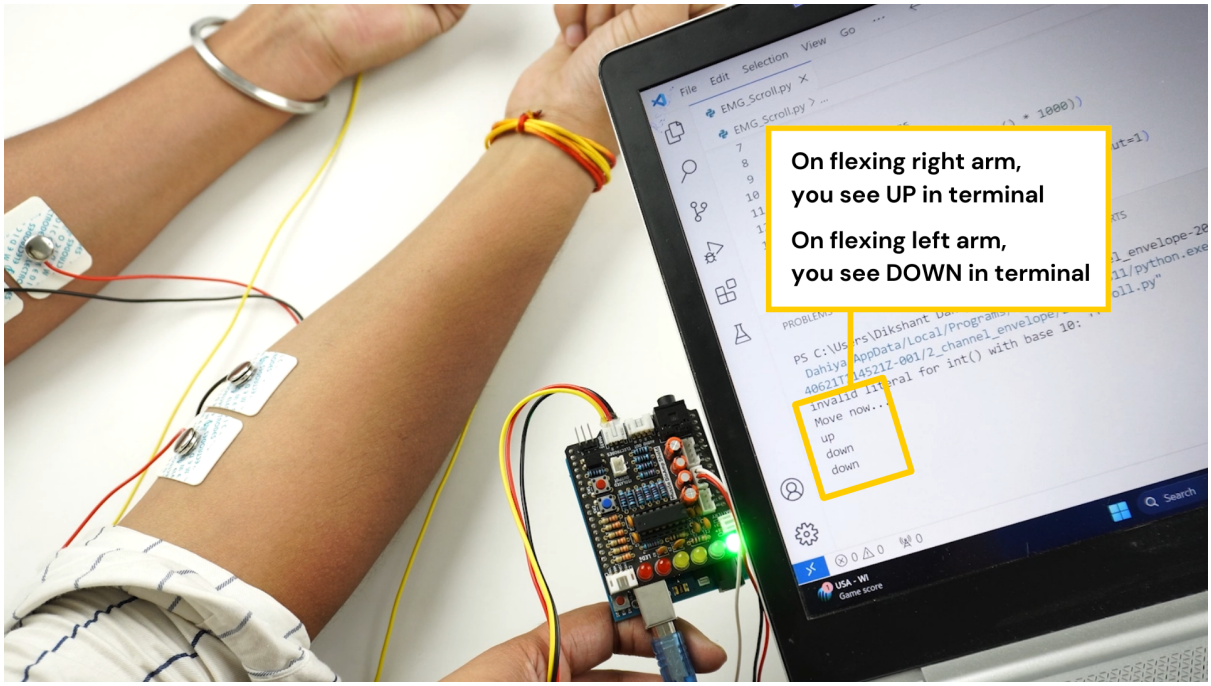
Press the `SW1 button` on Muscle BioAmp Shield again.

In the terminal, you will see Move Now prompt. When you flex your right arm, you'll see UP in the terminal. Similarly, when you move your left arm, you'll see DOWN in the terminal.

Now, open youtube shorts on your laptop and start scrolling using your muscle signals.

**Note**

What's happening in the background? Whenever an EMG signal is detected, it acts as a trigger to emulate UP or DOWN key on the keyboard.




### k. Calibrating the code

**Changes in Arduino Sketch**

Modify the threshold values on **lines 73 and 74**. Threshold 1 is for the EMG signals recorded from the Muscle BioAmp Shield, and threshold 2 is for the EMG signals recorded from the BioAmp EXG Pill.

Uncomment line 71 in the Arduino code and navigate to Tools > Serial Plotter. You'll see two plots showing the EMG signals of both of your arms. Flex your right arm and observe the peak value on the y-axis. If the peak value is around 240, you can set threshold 1 anywhere between 150 to 200. The lower the threshold value, the easier it is to trigger the output as UP or DOWN, and vice versa. Repeat the process to determine the threshold 2 value for your left arm.

After setting the thresholds, comment out line 71.

**Changes in Python script**

Adjust the latency value on **line 51**. A higher latency value will make the output less responsive, requiring you to flex and hold longer to scroll through the screen. A lower latency value will make the output more responsive, allowing you to scroll through the screen more easily.

#### I. Conclusion

This was just a demonstration to show you how both the sensors (BioAmp EXG Pill & Muscle BioAmp Shield) can be used together to create a 2-channel EXG acquisition system. In this project, we used BioAmp EXG Pill to record EMG signals, but it can also be used to record other biopotential signals as well like ECG, EOG, or EEG.

## 1.5 Some project ideas

You can find step-by-step tutorials for various HCI/BCI projects on our Instructables.

### 1.5.1 Projects made using BioAmp EXG Pill

1. Controlling video game using brainwaves (EEG)
2. Visualising electrical impulses from eyes (EOG)
3. Recording EEG from visual cortex part of brain
4. Recording EEG from prefrontal cortex part of brain
5. Eye blink detection
6. Creating a drowsiness detector
7. Record publication-grade ECG
8. Measuring heart rate
9. Detecting heart beats
10. Record publication-grade EMG
11. Detecting up and down movement of eyes

### 1.5.2 Projects made using Muscle BioAmp Shield

1. Record, visualise, and listen to EMG signals
2. Controlling 3d-printed servo claw using EMG
3. Controlling prosthetic hand using EMG
4. Building the ultimate servo claw game
5. Building muscle strength game

### 1.5.3 Projects made using the sensors together

1. Control dino game using eye blinks
2. Control servo claw using EOG
3. Visualize2-chEMGonLCDdisplaymodulehttps://www.instructables.com/Visualizing-2-Channel-EMG-on-LCD-Display-Mo
4. Control video games using 2-ch EMG

These are some of the project ideas but the possibilities are endless. So create your own Human Computer Interface (HCI) and Brain Computer Interface (BCI) projects and share them with us at `contact@upsidedownlabs.tech`

# 2 BioAmp EXG Pill

v1.0

## 2.1 Overview

BioAmp EXG Pill is a small, powerful analog-front-end (AFE) biopotential signal-acquisition board that can be paired with any microcontroller unit (MCU) or single-board computer (SBC) with an analog-to-digital converter (ADC) such as Arduino UNO & Nano, Adafruit QtPy, STM32 Blue Pill, BeagleBone Black, and Raspberry Pi Pico, to name just a few. It also works with any dedicated ADC, like the Texas Instruments ADS1115 and ADS131M0x, among others.

**Note**

It is recommended to use Arduino UNO R4 while recording biopotential signals since it has 14-bit ADC and can record the signals much accurately.

## 2.2 What makes it different?

1. Record publication-quality biopotential signals like ECG, EMG, EOG, or EEG.
2. Small size (25.4 x 10.0mm) allows easy integration into mobile and space-constrained projects.
3. Powerful noise rejection makes it usable even when the device is close to the AC mains supply.
4. Any 1.5 mm diameter wire can be used as a strain-relieving electrode cable, making it very cost-effective.
5. Pair it with any MCU with an ADC. It is by default compatible with 5V but you can make it compatible with 3.3V as well using a voltage divider.
6. Configure the gain, band pass filter and electrode count according to your requirements.

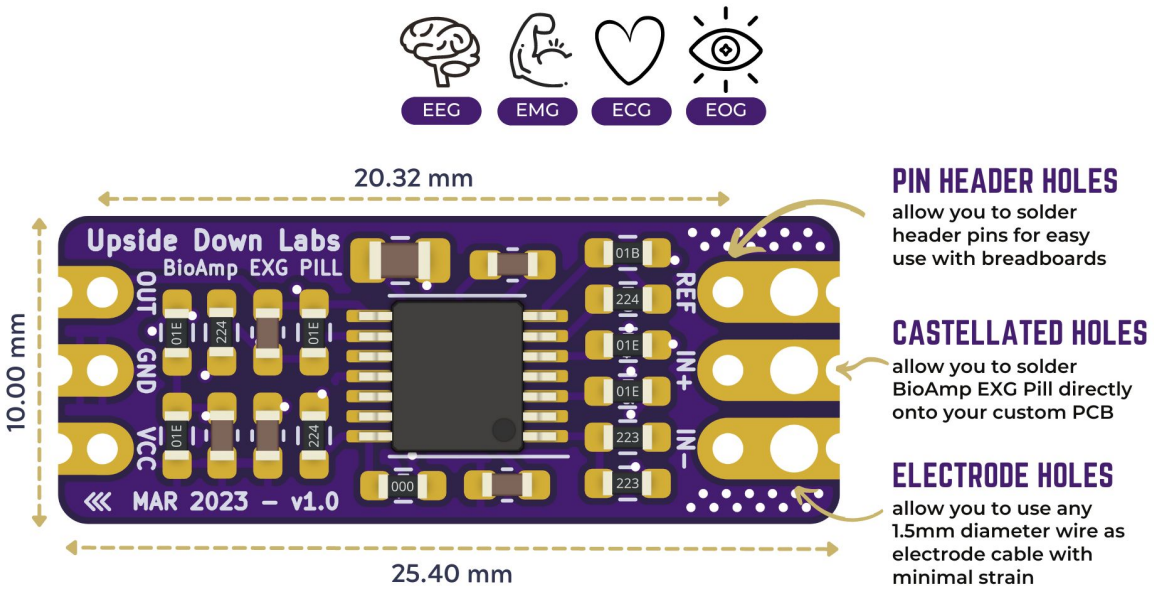
## 2.3 Features & Specifications

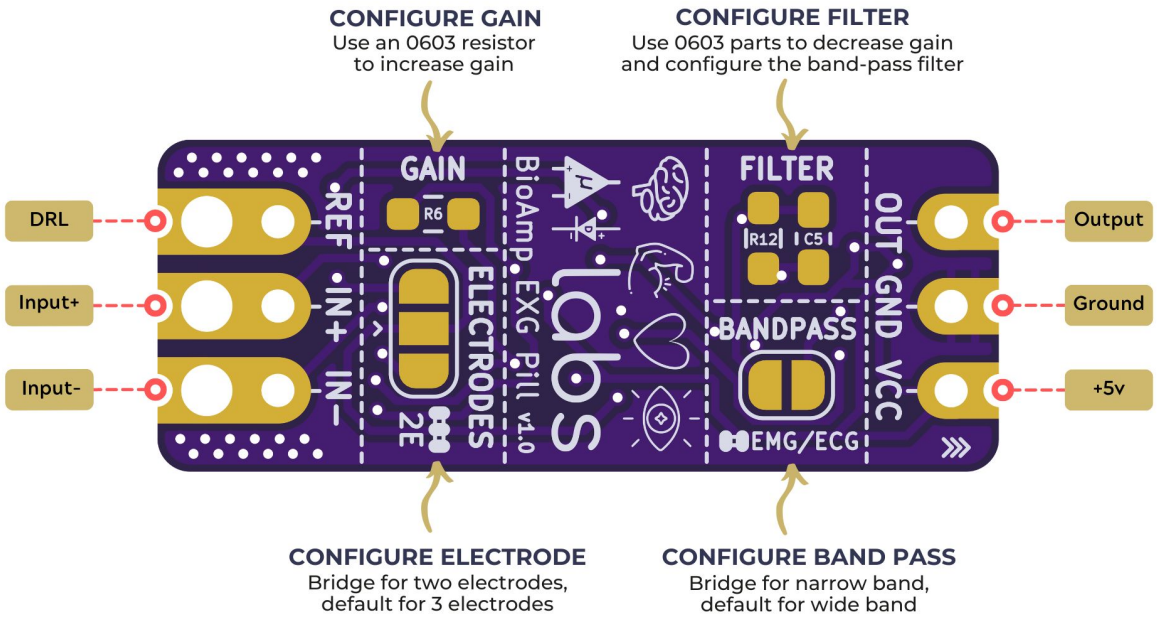
| | |
|---|---|
| Operating Voltage | 3.3 V / 5 V |
| Input Impedance | 10^12 ohm |
| Compatible Hardware | Any development board with an ADC (Arduino UNO & Nano, Adafruit QtPy, STM32 Blue Pill, BeagleBone Black, Raspberry Pi Pico, to name just a few) or any standalone ADC of your choice |
| Biopotential Signals | EMG, ECG, EOG, EEG (configurable band-pass, by default configured for EEG & EOG) |
| No. of channels | 1 |
| Electrodes | 2 or 3 (By default configured for 3 electrodes) |
| Dimensions | 25.4 x 10 mm |
| Open Source | Hardware + Software |

## 2.4 Board layout

BioAmp EXG Pill's elegant design allows it to be used in 3 ways:

1. Pin-header holes allow you to solder (berg strip) pin headers for easy use with a breadboard.
2. Castellated holes allow you to solder BioAmp EXG Pill directly onto a custom PCB that requires biopotential-amplification capabilities.
3. Electrode holes allow you to use any 1.5 mm diameter wire as an electrode cable with minimal strain.


EEG
EMG
ECG
EOG
20.32 mm
10.00 mm
25.40 mm
Upside Down Labs
BioAmp EXG PILL
OUT
GND
VCC
REF
IN+
IN−
MAR 2023 – v1.0
PIN HEADER HOLES
allow you to solder header pins for easy use with breadboards
CASTELLATED HOLES
allow you to solder BioAmp EXG Pill directly onto your custom PCB
ELECTRODE HOLES
allow you to use any 1.5mm diameter wire as electrode cable with minimal strain

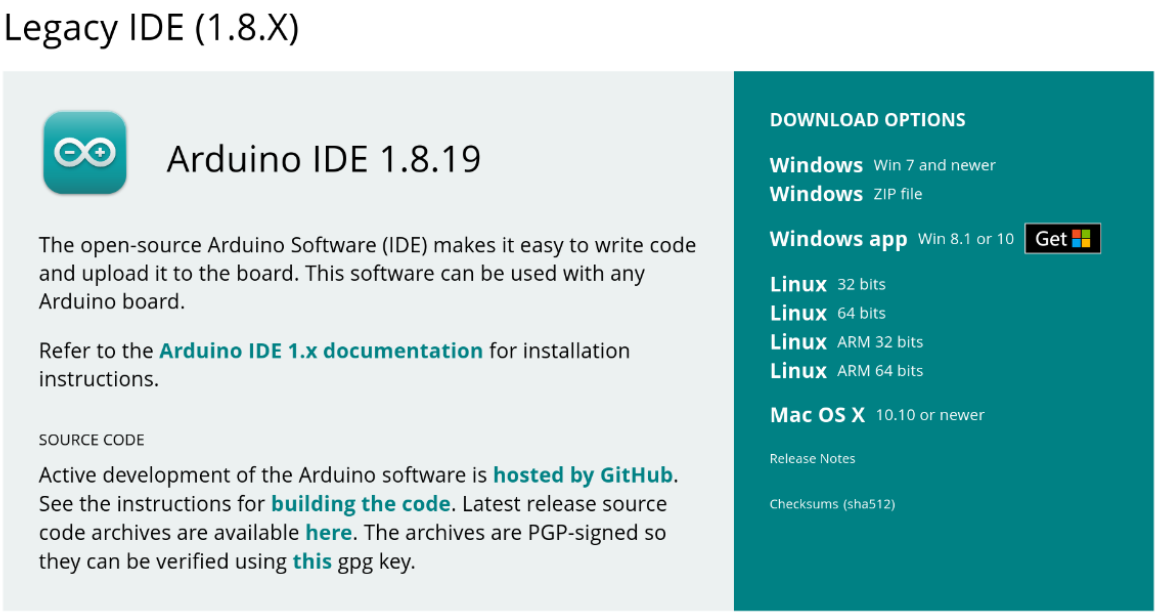

CONFIGURE GAIN
Use an 0603 resistor to increase gain
CONFIGURE FILTER
Use 0603 parts to decrease gain and configure the band-pass filter
DRL
Input+
Input-
REF
IN+
IN−
GAIN
R6
ELECTRODES
2E
BioAmp EXG Pill v1.0
labs
FILTER
R12
C5
BANDPASS
EMG/ECG
OUT
GND
VCC
Output
Ground
+5v
CONFIGURE ELECTRODE
Bridge for two electrodes, default for 3 electrodes
CONFIGURE BAND PASS
Bridge for narrow band, default for wide band

### 2.4.1 BioAmp EXG Pill is fully configurable

1. Increase the gain of the instrumentation amplifier by using a 0603 resistor at R6. Decrease gain and configure the bandpass filter by using 0603 parts at R12 and C5. Band limiting is very useful for EOG and EEG recording. Also, the signal sometimes clips while recording an ECG with electrodes very close to the heart. Creating a solder jumper for a band-pass filter helps with that. By default, BioAmp EXG Pill is configured to record EEG and EOG but you can bridge the pads (below bandpass) with solder to make it configurable for EMG and ECG.
2. The normal method of operation for best-quality signal amplification is to use 3 electrodes by default but you can bridge the pads (below electrodes) to make it configurable for 2 electrodes. The 2-electrode mode is specifically included for projects like heart (ECG) patches for HRV. It's only supposed to be used with a battery-operated setup and is quite prone to high interference noise due to a lack of proper reference on the body (This option is not recommended for most operations)

## 2.5 Software Requirements

1. **Arduino IDE**
   - Latest Version (v2.3.8+): Arduino IDE Latest Version Download *(Recommended for faster compilation and enhanced support for the latest Arduino UNO Development Boards)*
   - Legacy Version (v1.8.19): Arduino IDE v1.8.19 (legacy IDE) Download *(For Real-time signal visualization using a better built-in Serial Plotter)*




2. **Chords Web Visualizer**

   Use Chords Web, our custom web interface designed to visualize biopotential signals data directly in your browser. *(Plug-and-play signal visualization directly in any modern web browser.)* Flash the Chords-Arduino-Firmware on your board to visualize signals on Chords Web.

> **Note**
>
> For detailed instructions and visualization tips, check the Chords Web Documentation.
>
> https://youtu.be/IVIPnk9z75g

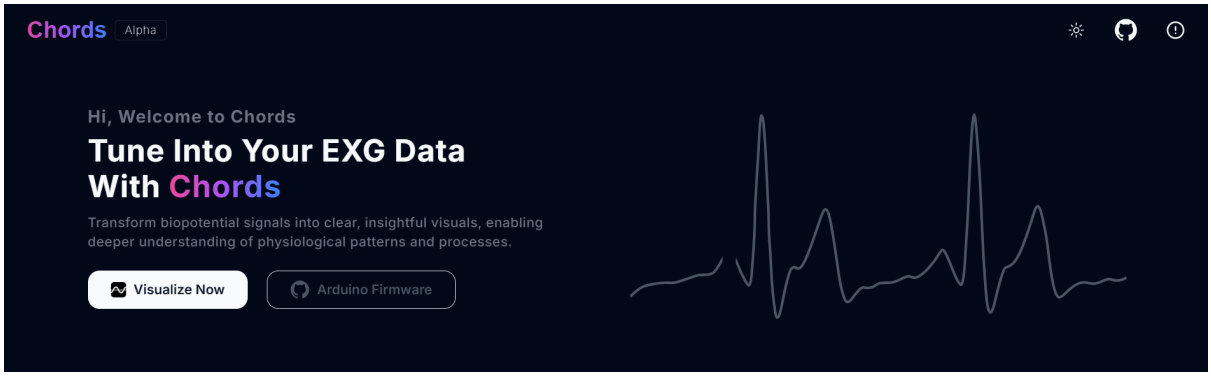



Fig. 1: Chords Web Landing Page

## 2.6 Using the Hardware

If you have received the assembled BioAmp EXG Pill then you can skip the step 1 and move on to step 2.

### 2.6.1 Step 1: Solder Connectors

Insert the provided BioAmp cable's JST PH connector and header pins from top as shown in the image and solder them from below.

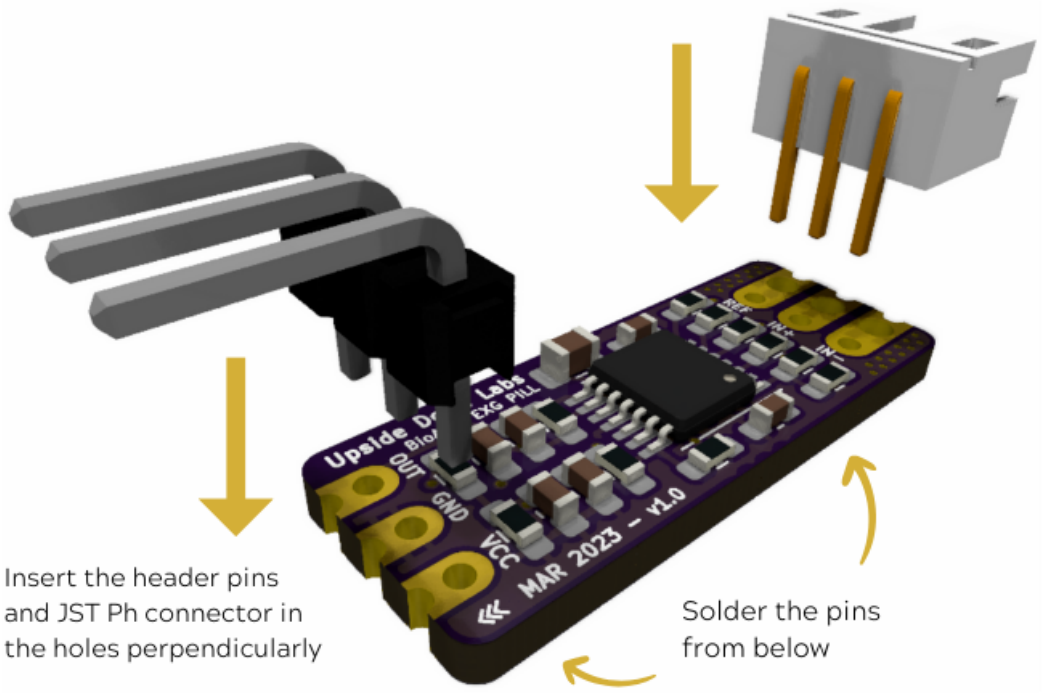



Fig. 2: Soldering the connector & header pins on BioAmp EXG Pill

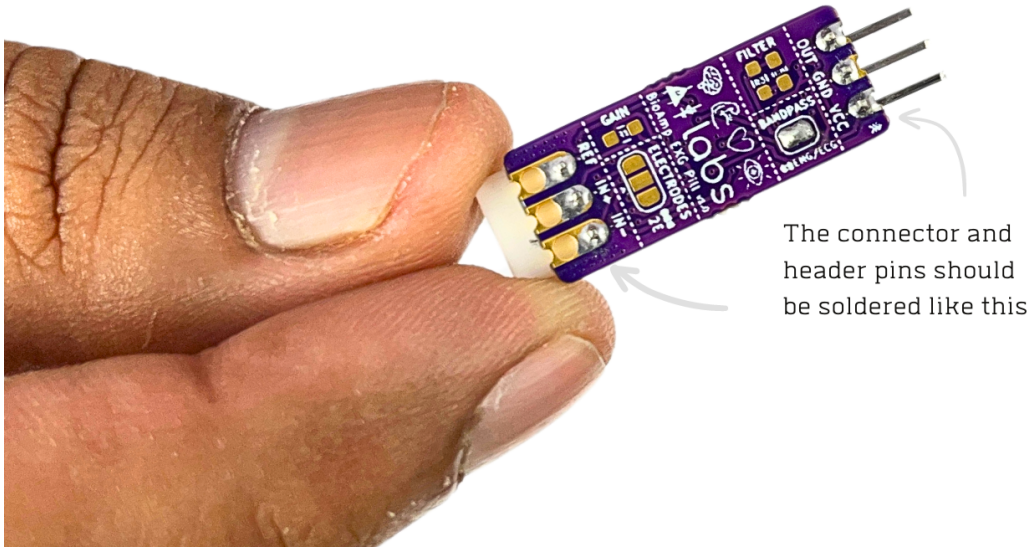
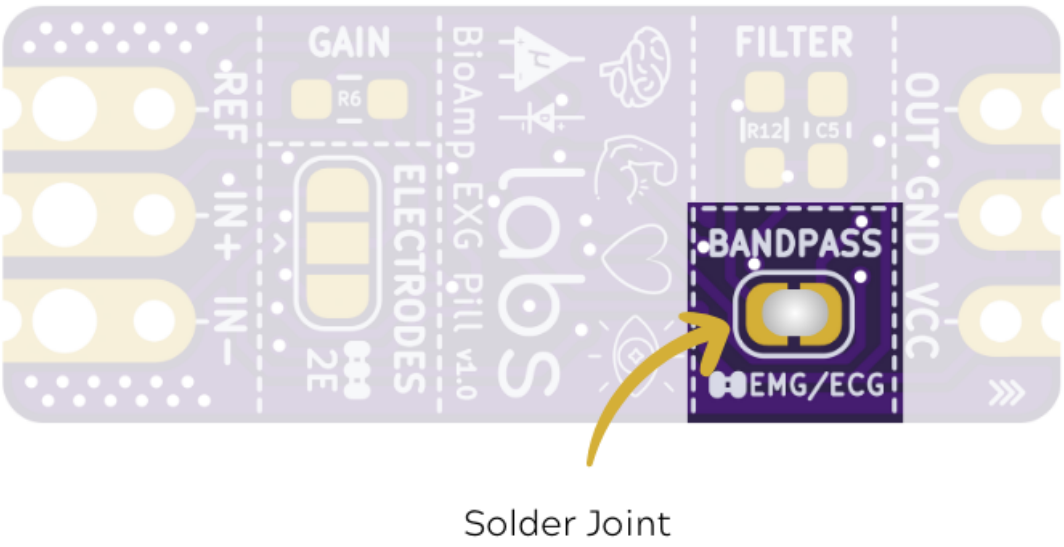



Fig. 3: After soldering, BioAmp EXG Pill should look like this

### 2.6.2 Step 2 (optional): Configure for ECG/EMG

BioAmp EXG Pill is by default configured for recording EEG or EOG but if you want to record good quality ECG or EMG, then it is recommended to configure it by making a solder joint as shown in the image.




**Note**

Even without making the solder joint the BioAmp EXG Pill is capable of recording ECG or EMG but the signals would be more accurate if you configure it.

### 2.6.3 Step 3: Connect MCU/ADC

Connect your BioAmp EXG Pill to your MCU/ADC as per the connection table shown below:

Table 1: BioAmp to MCU/ADC connection

| BioAmp EXG Pill | MCU/ADC | Function |
|---|---|---|
| VCC | 5V / 3.3V | Power Supply |
| GND | GND | Ground |
| OUT | ADC Input | Signal |

For all the examples provided, we are using the A0 pin of Arduino UNO R3. Connect your BioAmp to your MCU/ADC via jumper cables provided in the kit. If you are connecting `OUT` pin of BioAmp to any other analog pin (A0-A5) of Arduino UNO board, then you will have to change the INPUT PIN in the Arduino sketch accordingly.

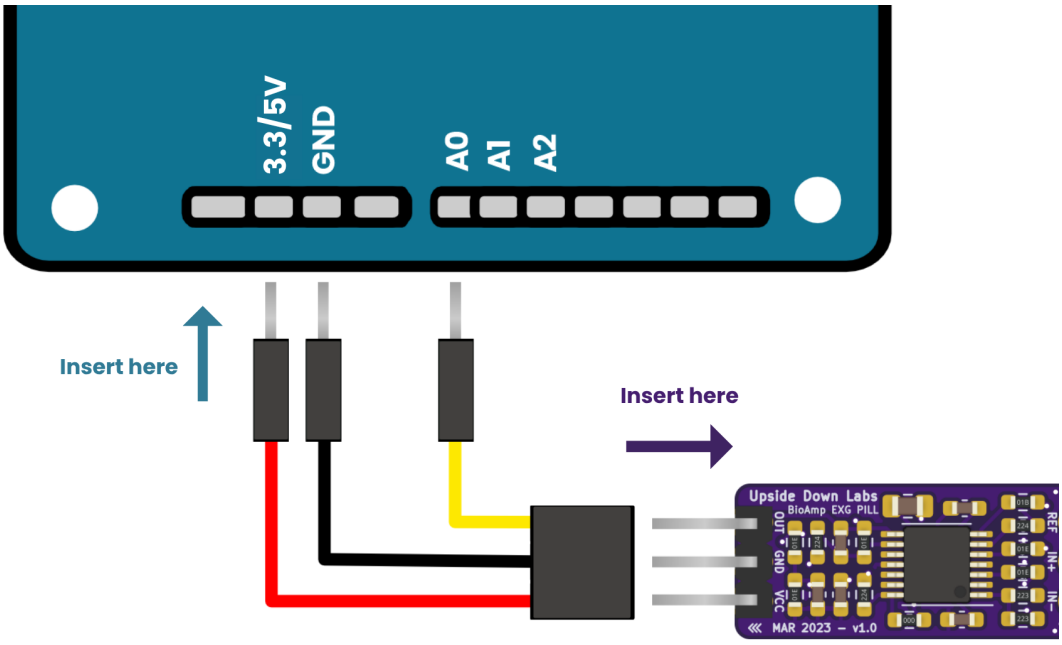



Fig. 4: Wiring diagram for VCC, GND, and OUT Connections with Arduino UNO

**Warning**

Take precautions while connecting to power, if power pins are to be swapped, your BioAmp EXG Pill will be fried and it'll become unusable (DIE).

### 2.6.4 Step 4: Connecting electrode cable

Connect the BioAmp cable to BioAmp EXG Pill by inserting the cable end in the JST PH connector as shown in the graphic below.

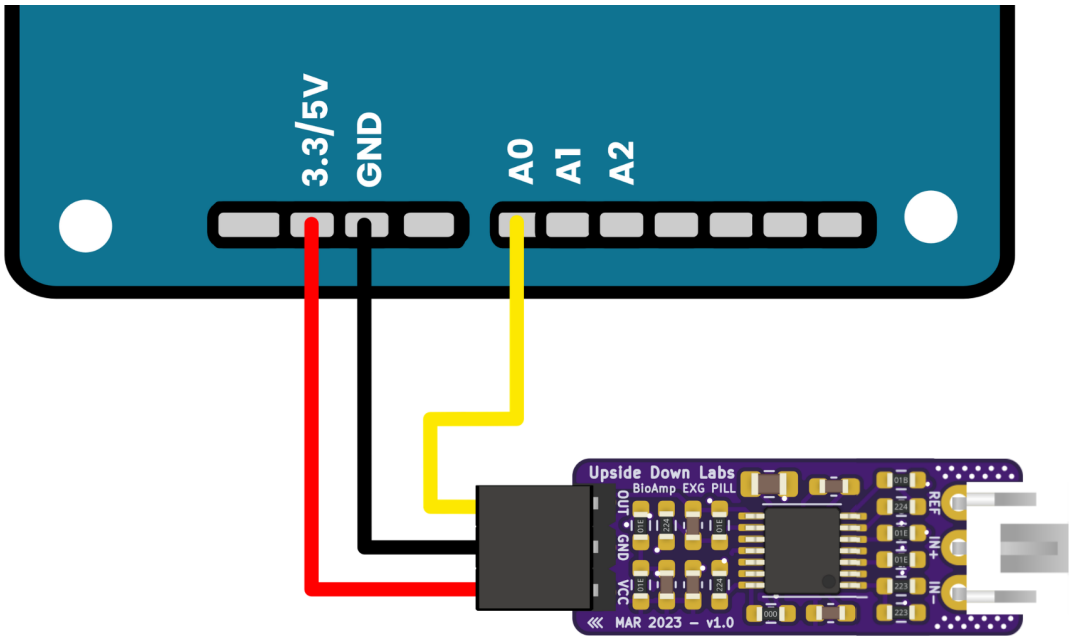



Fig. 5: Connections with Arduino UNO Development Boards

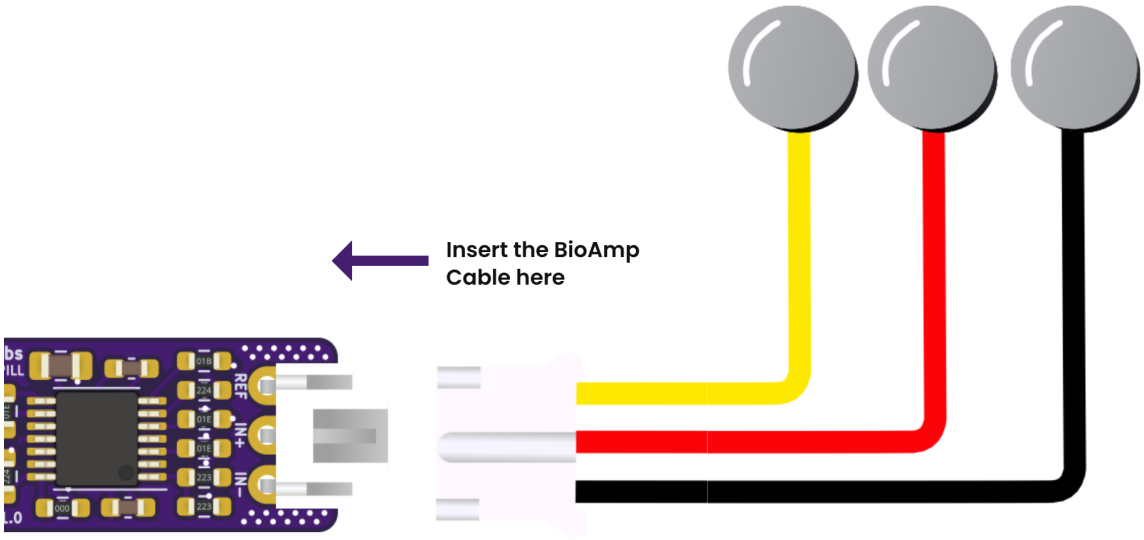



Fig. 6: Connecting the BioAmp Cable v3 to the BioAmp EXG Pill

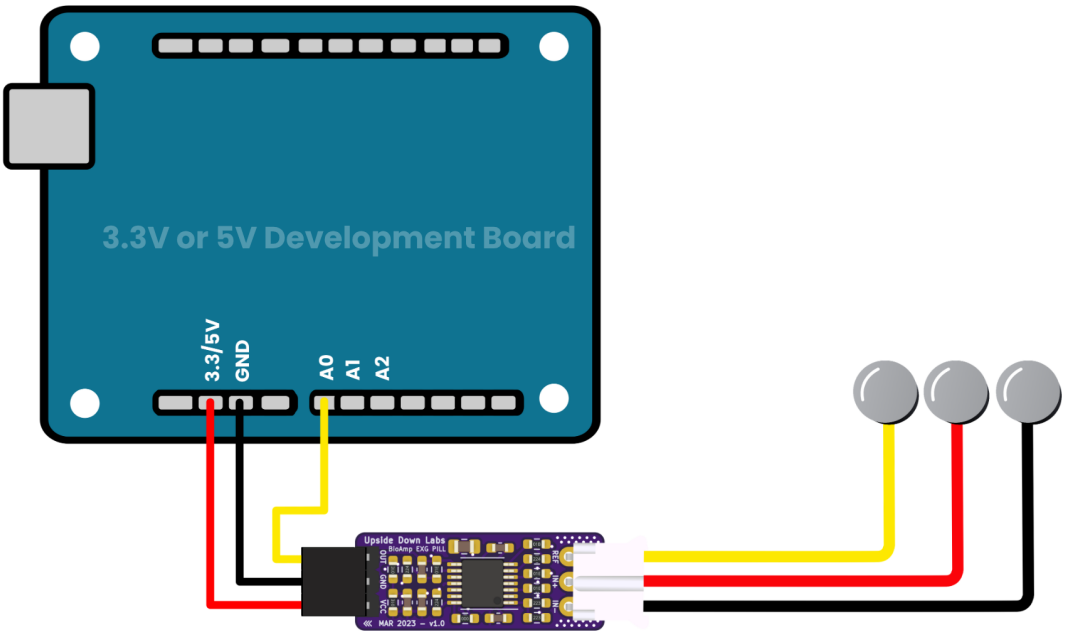



Fig. 7: Complete assembly of BioAmp EXG Pill, Arduino, and BioAmp Cable v3

### 2.6.5 Step 5: Skin Preparation

Apply Nuprep Skin Preparation Gel on the skin surface where electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step Skin Preparation Guide.

### 2.6.6 Step 6: Measuring ElectroMyoGraphy (EMG)

**Note**

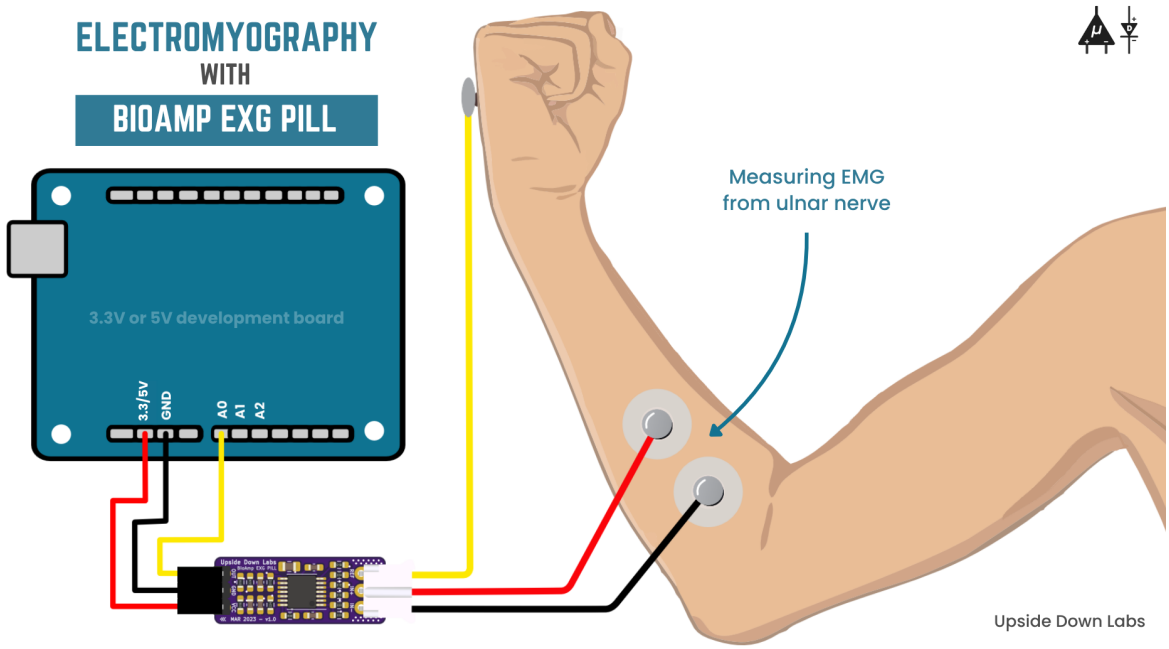
- Electromyography (EMG) is a technique for evaluating and recording the electrical activity produced by skeletal muscles.
- EMG is also used as a diagnostic procedure to assess the health of muscles and the nerve cells that control them (motor neurons).
- EMG results can reveal nerve dysfunction, muscle dysfunction, or problems with nerve-to-muscle signal transmission.

#### Electrodes placement

We have 2 options to measure the EMG signals, either using the gel electrodes or using dry electrode based Muscle BioAmp Band. You can try both of them one by one.

- **Using Gel electrodes:**

1. Connect the BioAmp cable to gel electrodes,
2. Peel the plastic backing from electrodes
3. Place the IN+ and IN- cables on the arm near the ulnar nerve & REF (reference) at the back of your hand as shown in the connection diagram.




- **Using Muscle BioAmp Band:**

1. Connect the BioAmp cable to Muscle BioAmp Band in a way such that IN+ and IN- are placed on the arm near the ulnar nerve & REF (reference) on the far side of the band.
2. Now put a small drop of electrode gel between the skin and metallic part of BioAmp cable to get the best results.

**Tip**

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

**Tutorial on how to use the band:**

https://youtu.be/xYZdw0aesa0

**Note**

In this demonstration we are recording EMG signals from the ulnar nerve, but you can record EMG from other areas as well (biceps, triceps, legs, jaw etc) as per your project requirements. Just make sure to place the IN+, IN- electrodes on the targeted muscle and REF on a bony part.

### Uploading the code

- Connect the Arduino UNO Development Board to your laptop using the USB cable. To upload the code use Arduino IDE latest version (v2.3.8+).
- Choose and copy one of the following Arduino sketches based on your requirements:

EMG Filter

EMG Envelope

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).

- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

**Tip**

Check our complete documentation on How to upload the code.

**Warning**

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

### Visualizing the EMG signals

1. **Using Chords Web**

   For visualizing the EMG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

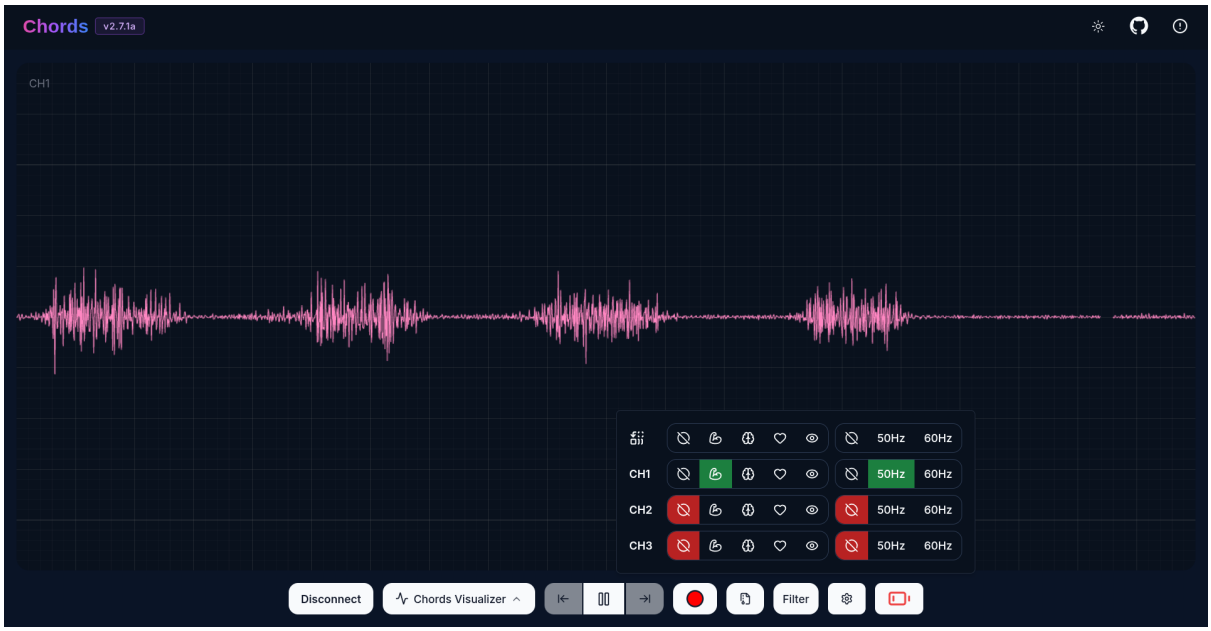



Fig. 8: Visualizing EMG signals on Chords Web

Now flex your arm to visualize the muscle signals in real time on your laptop.

1. **Using Arduino legacy v1.8.19**

   To visualize the signal on Arduino v1.8.19 we will use inbuilt Serial Plotter:

- Navigate to (`Tools` > `Board`) and select your Development board or Arduino UNO Board (if not detected automatically).
- Open the Serial Plotter (`Tools` > `Serial Plotter` or `Ctrl`+`Shift`+`L`) to visualize the EMG signals.
- Once the Serial Plotter is open, ensure the baud rate is set to `115200`.
- Now flex your arm to visualize the muscle signals in real time on your laptop.

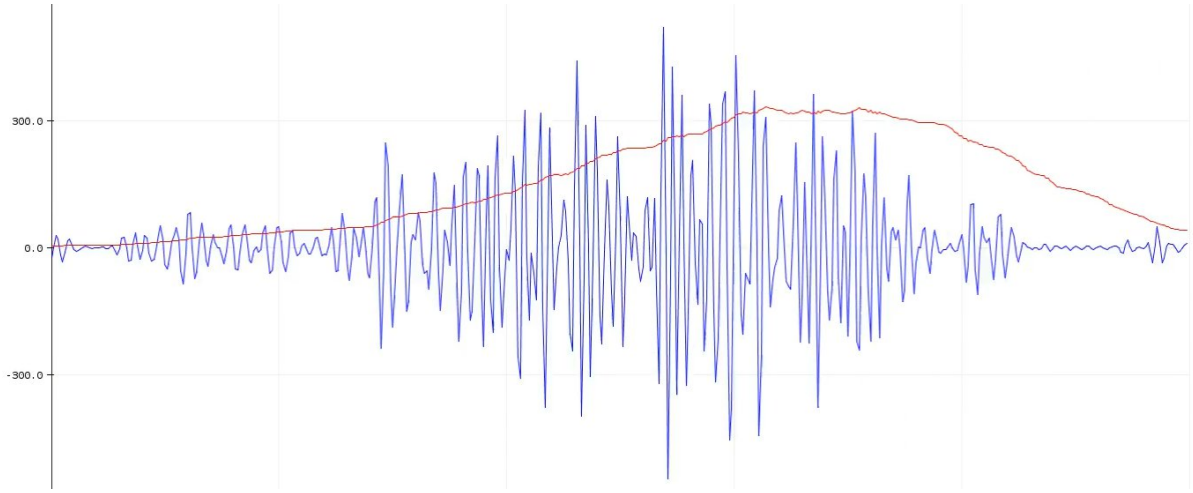



Fig. 9: Visualizing EMG signals on Arduino IDE v1.8.x

### 2.6.7 Step 6: Measuring ElectroCardioGraphy (ECG)

**Note**

- Electrocardiography (ECG) is the process of producing an electrocardiogram (ECG or EKG).
- It is a graph of voltage versus time of the electrical activity of the heart using electrodes placed on the skin.
- These electrodes detect the small electrical changes that are a consequence of cardiac muscle depolarization followed by repolarization during each cardiac cycle (heartbeat).

#### Electrodes placement

We have 2 options to measure the ECG signals, either using the gel electrodes or using dry electrode based Heart BioAmp Band. You can try both of them one by one.

- **Using Gel electrodes:**

1. Connect the BioAmp cable to gel electrodes
2. Peel the plastic backing from electrodes
3. Place the IN- cable on the left side, IN+ in the middle and REF (reference) on the far right side as shown in the diagram.

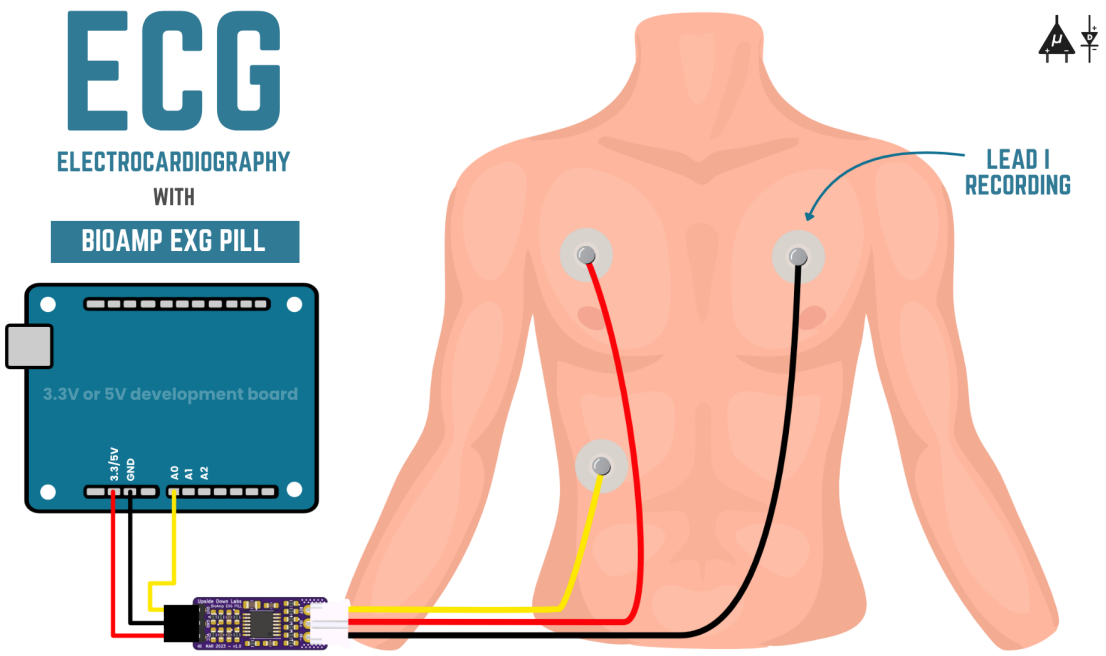
- **Using Heart BioAmp Band:**

1. Wear the band as depicted in the video tutorial given below
2. Place the IN- cable on the left side, IN+ in the middle and REF (reference) on the far right side.
3. Now put a small drop of electrode gel between the skin and metallic part of BioAmp cable to get the best results.

**Tip**

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

**Tutorial on how to use the band:**




https://youtu.be/fr5iORsVyUM

### Uploading the code

- Connect the Arduino UNO Development Board to your laptop using the USB cable. To upload the code use Arduino IDE latest version (v2.3.8+).
- Choose and copy one of the following Arduino sketches based on your requirements:

ECG Filter

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).
- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

**Tip**

Check our complete documentation on How to upload the code.

**Warning**

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

### Visualizing the ECG signals

1. **Using Chords Web**

   For visualizing the ECG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

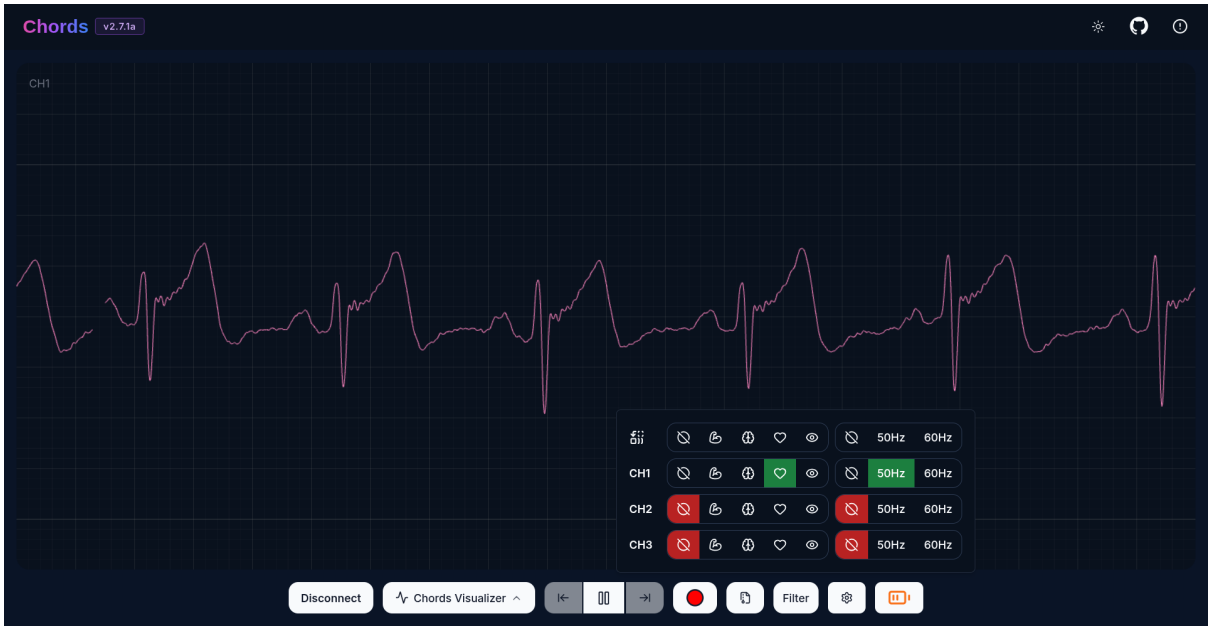



Fig. 10: Visualizing ECG signals on Chords Web

Sit back, relax and see your ECG signals in real time on your laptop.

2. **Using Arduino legacy v1.8.19**

   To visualize the signal on Arduino v1.8.19 we will use inbuilt Serial Plotter:

- Navigate to (`Tools` > `Board`) and select your Development board or Arduino UNO Board (if not detected automatically).
- Open the Serial Plotter (`Tools` > `Serial Plotter` or `Ctrl`+`Shift`+L) to visualize the EMG signals.
- Once the Serial Plotter is open, ensure the baud rate is set to `115200`.
- Now sit back, relax and see your ECG signals in real time on your laptop.

## 2.6.8 Step 7: Measuring Electrooculography (EOG)

**Note**

- Electrooculography (EOG) is a technique for measuring the corneo-retinal standing potential that exists between the front and the back of the human eye. The resulting signal is called EOG.
- To measure eye movement, pairs of electrodes are typically placed either above and below the eye or to the left and right of the eye.
- If the eye moves from the center position toward one of the two electrodes, this electrode "sees" the positive side of the retina, and the opposite electrode "sees" the negative side of the retina.
- Consequently, a potential difference occurs between the electrodes. Assuming the resting potential is constant, the recorded potential is a measure of the eye's position.

Fig. 11: Visualizing ECG signals on Arduino IDE v1.8.x

### Electrodes placement

We have 2 ways to measure the EOG signals, either record the horizontal eye movement or the vertical eye movement. You can one by one record both the signals.

- **Horizontal EOG recording:**

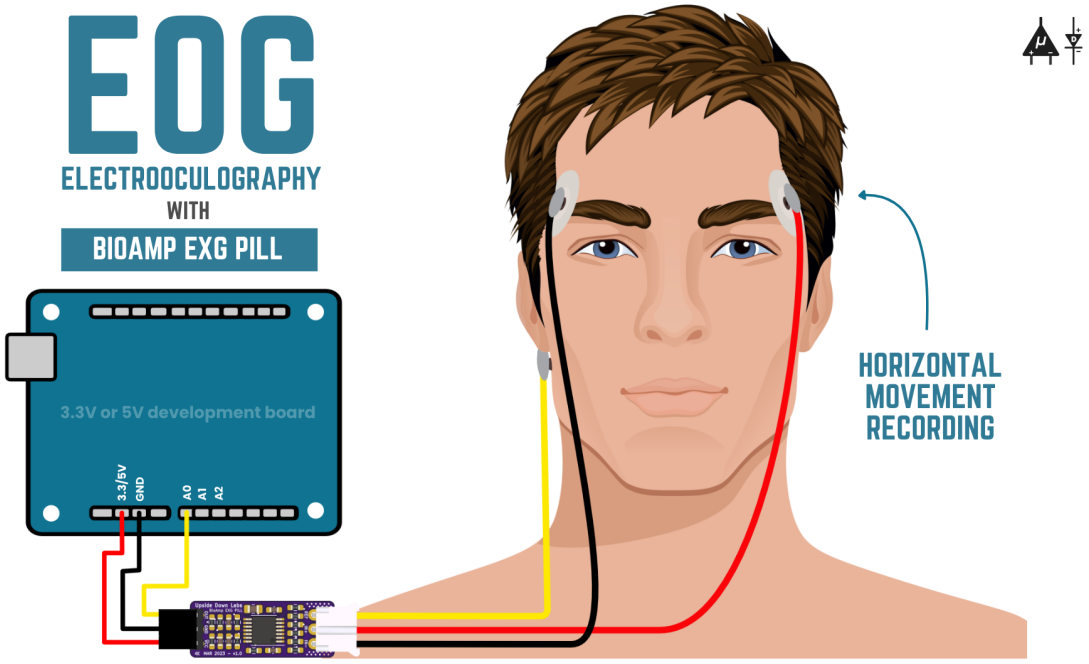



Fig. 12: Horizontal EOG Placement

1. Connect the BioAmp cable to gel electrodes.
2. Peel the plastic backing from electrodes.
3. Place the IN- cable on the right side of the eye, IN+ on the left side of the eye and REF (reference) at the bony part, on the back side of your earlobe as shown in the diagram above.

- **Vertical EOG recording:**

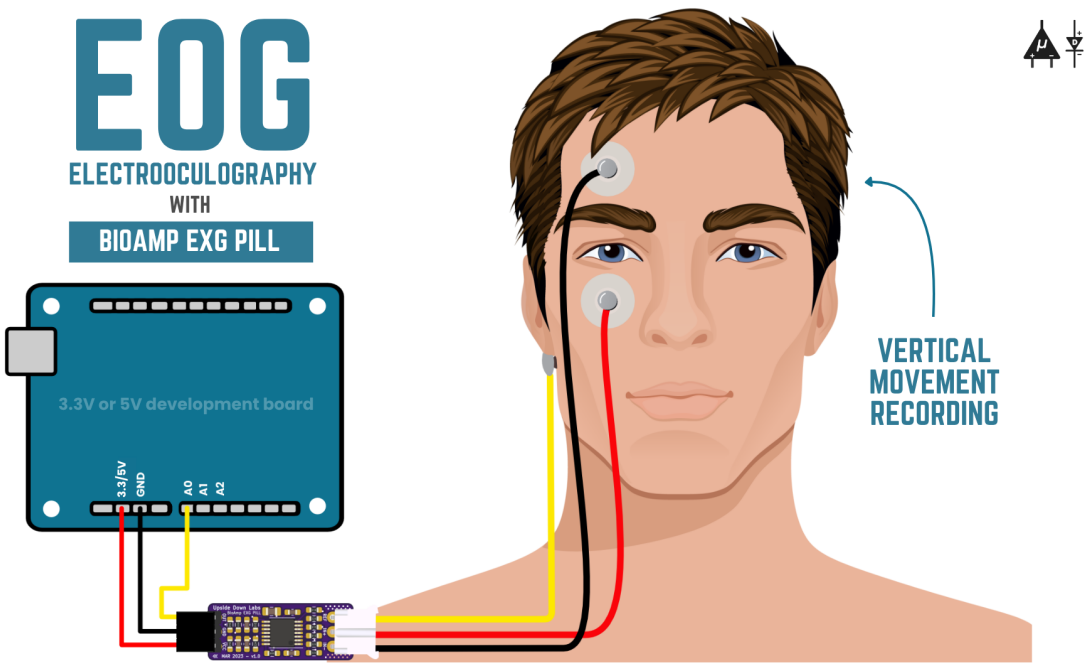



Fig. 13: Vertical EOG Placement

1. Connect the BioAmp cable to gel electrodes.
2. Peel the plastic backing from electrodes.
3. Place the IN- & IN+ cables above and below the eye respectively and REF (reference) at the bony part, on the back side of your earlobe as shown in the diagram above.

### Uploading the code

- Connect the Arduino UNO Development Board to your laptop using the USB cable. To upload the code use Arduino IDE latest version (v2.3.8+).
- Choose and copy one of the following Arduino sketches based on your requirements:

EOG Filter

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).
- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

**Tip**

Check our complete documentation on How to upload the code.

**Warning**

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

### Visualizing the EOG signals

1. **Using Chords Web**

   For visualizing the EOG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

Fig. 14: Visualizing EOG signals on Chords Web

Move your eyes up-down or left-right to see your EOG signals in real time on your laptop.

2. **Using Arduino legacy v1.8.19**

   To visualize the signal on Arduino v1.8.19 we will use inbuilt Serial Plotter:

- Navigate to (`Tools` > `Board`) and select your Development board or Arduino UNO Board (if not detected automatically).
- Open the Serial Plotter (`Tools` > `Serial Plotter` or `Ctrl`+`Shift`+L) to visualize the EMG signals.
- Once the Serial Plotter is open, ensure the baud rate is set to `115200`.
- Now move your eyes up-down or left-right to see your EOG signals in real time on your laptop.

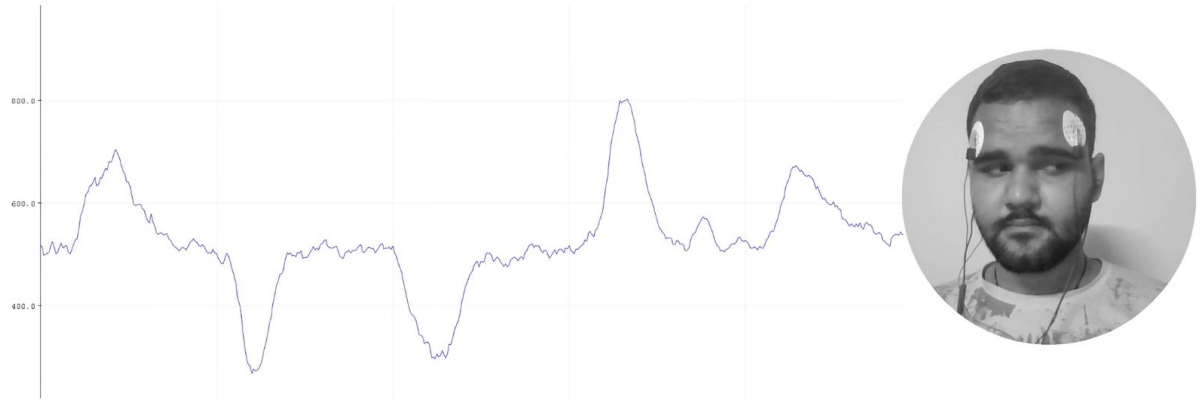



Fig. 15: Visualizing EOG signals on Arduino IDE v1.8.x

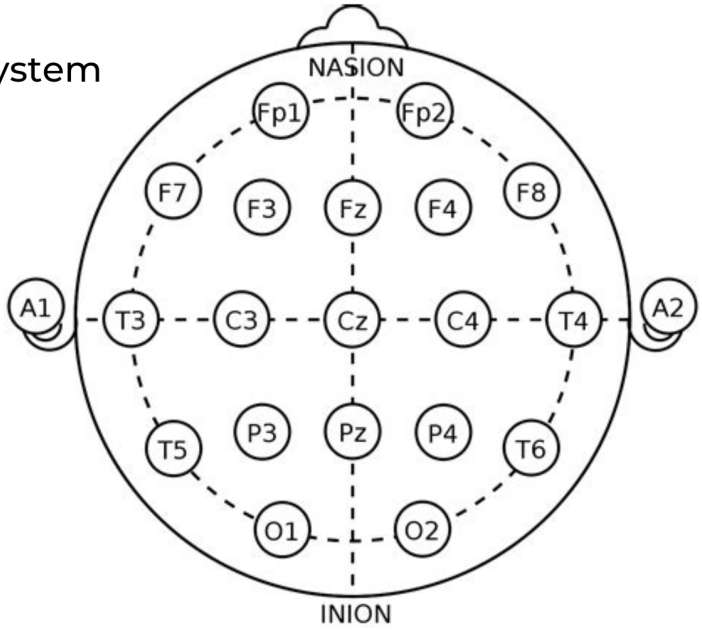
### 2.6.9 Step 8: Measuring Electroencephalography (EEG)

**Note**

Electroencephalography (EEG) is an electrophysiological monitoring method to record electrical activity on the scalp. During the procedure, electrodes consisting of small metal discs with thin wires are pasted onto your scalp. The electrodes detect tiny electrical charges that result from the activity of your brain cells which are then amplified to appear on the computer screen. It is typically non-invasive, with the electrodes placed along the scalp.

For recording EEG from different parts of the brain, you have to place the electrodes according to the International 10-20 system for recording EEG.

International 10-20 system for recording EEG




Source: Wikipedia

#### Electrodes placement

We have 2 options to measure the EEG signals, either using the gel electrodes or using dry electrode based Brain BioAmp Band. You can try both of them one by one.

- **Using gel electrodes to record from prefrontal cortex part of brain:**

1. Connect the BioAmp cable to gel electrodes.
2. Peel the plastic backing from electrodes.
3. Place the IN+ and IN- cables on Fp1 and Fp2 as per the International 10-20 system & REF (reference) at the bony part, on the back side of your earlobe as shown above.

- **Using Brain BioAmp Band to record from prefrontal cortex part of brain:**

1. Connect the BioAmp cable to Brain BioAmp Band in a way such that IN+ and IN- are placed on Fp1 and Fp2 as per the International 10-20 system.
2. In this case, the REF (reference) should be connected using gel electrode. So connect the reference of BioAmp cable to the gel electrode, peel the plastic backing and place it at the bony part, on the back side of your earlobe.

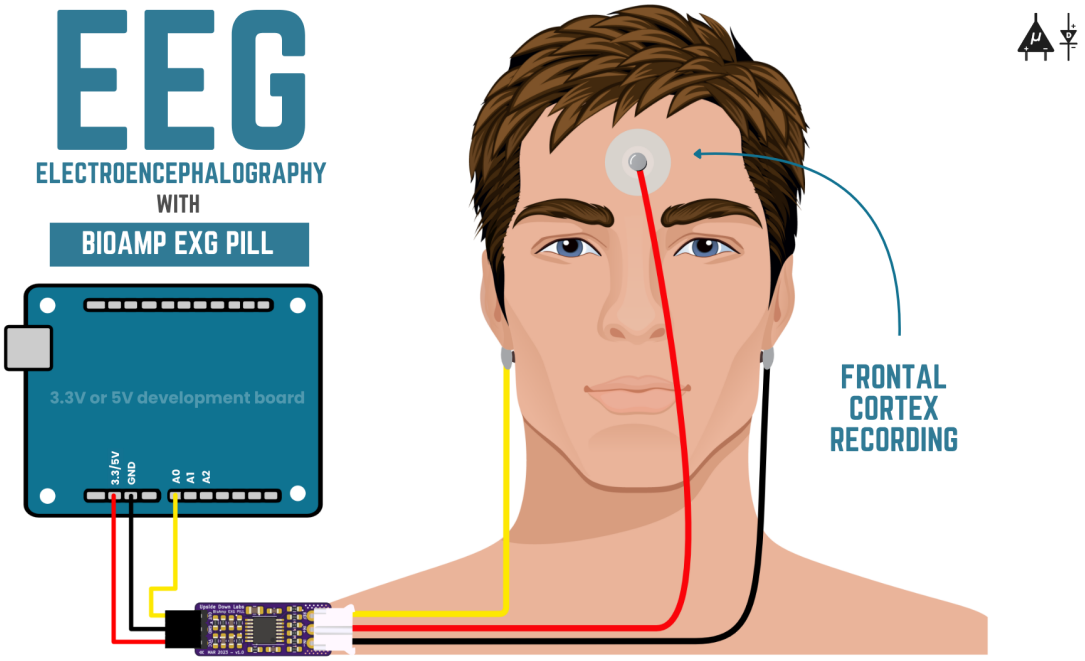



Fig. 16: EEG Placement

3. Now put a small drop of electrode gel on the dry electrodes (IN+ and IN-) between the skin and metallic part of BioAmp cable to get the best results.

**Tip**

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

**Tutorial on how to use the band:**

https://youtu.be/O6qp7teT-sM

**Note**

Similarly you can use the band to record EEG signals from the visual cortex part of brain by placing the dry electrodes on O1 and O2 instead of Fp1 and Fp2. Everything else will remain the same.

### 2-Channel EEG data

To record 2-channel Electroencephalography (EEG) data using two BioAmp EXG Pills in conjunction with an Arduino Uno R4 Minima, following hardware setup is essential.

**Required Components**

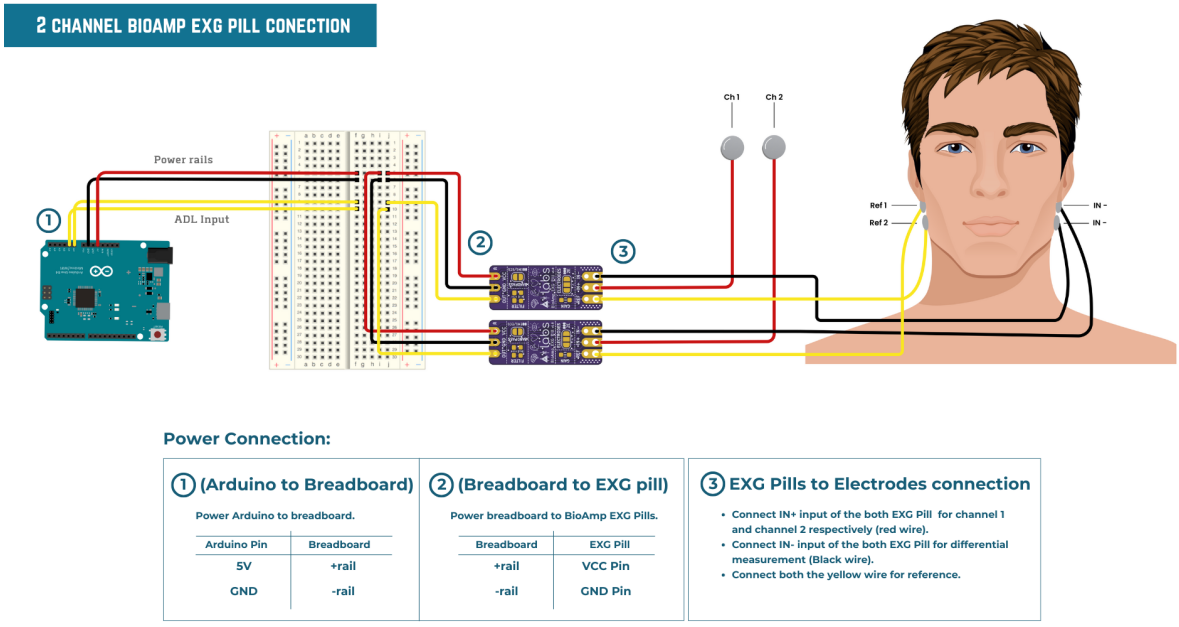
- 2 x BioAmp EXG Pill
- 1 x Arduino Uno R4 Minima
- 1 x Breadboard
- Jumper wires
- Gel Electrodes

### Hardware setup

To set up the hardware for 2-channel EEG recording, follow the connection steps below:

1. **Powering the Modules:** Connect the Arduino's `5V` and `GND` pins to the power rails of the breadboard. Supply `VCC` and `GND` from the breadboard rails to both BioAmp EXG Pills.
2. **Signal Connections:** Connect the `OUT` pin of the first EXG Pill to the Arduino's analog pin `A0` (Channel 1). Connect the `OUT` pin of the second EXG Pill to analog pin `A1` (Channel 2).
3. **Electrode Placement:**
    - **Active Sites:** Connect the `IN+` (red wires) of each module to different EEG recording sites on the forehead (e.g., **Fp1** and **Fp2**).
    - **Reference & Ground:** Connect the `IN-` (black wires) and both `REF` (yellow wires) from both EXG Pills to a common reference electrode placed on a neutral location, such as the mastoid or earlobe.

This common reference configuration ensures clean and synchronized signal acquisition across both channels.




## Uploading the code

- Connect the Arduino UNO Development Board to your laptop using the USB cable. To upload the code use Arduino IDE latest version (v2.3.8+).
- Choose and copy one of the following Arduino sketches based on your requirements:

EEG Filter Code

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).
- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

**Tip**

Check our complete documentation on How to upload the code.

**Warning**

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

## Visualizing the EEG signals

### Using Chords Web

For visualizing the EEG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

Fig. 17: Visualizing EEG signals on Chords Web

The signals that you can see on the screen right now are originating from prefrontal cortex part of your brain and propagating through all the layers to the surface of your skin.

You have placed the electrodes on the forehead (Fp1 & Fp2), the BioAmp EXG Pill is amplifying those signals so that we can detect it and finally sending it to the ADC (Analog to Digital Convertor) of your Arduino Uno and the signals are being visualized in Chords Web.

**Tip**

To ensure you're recording a high-quality signal, refer to the detailed guide here: Troubleshooting EEG Signal Quality.

We hope everything is clear now and you understand how the signals are propagating from your brain to the screen of the laptop.

## 2.7 Glimpses of previous versions

The BioAmp EXG Pill can be used in a variety of ways, the YouTube video below shows a potential way of using `v0.7` of BioAmp EXG Pill.

https://youtu.be/-G3z9fvQnuw

A lot has improved in terms of interference rejection and flexibility from `v0.7` to `v1.0` of the BioAmp EXG Pill. The YouTube video below shows the ECG, EMG, EOG, and EEG recording using `v1.0b` of device.

https://youtu.be/z9-B9bHWuhg

## 2.8 Real-world Applications

BioAmp EXG Pill is perfect for researchers, makers, and hobbyists looking for novel ways to sample biopotential data. It can be used for a wide variety of interesting biosensing projects, including:

- AI-assisted detection of congestive heart failure using CNN (ECG)
- Heart-rate variability calculation to detect heart ailments (ECG)
- Prosthetic arm (servo) control (EMG)
- Controlling a 3DOF robotic arm (EMG)
- Real-time game controllers (EOG)
- Blink detection (EOG)
- Capturing photos with a blink of an eye (EOG) and many more examples.

## 2.9 Project ideas & tutorials

You can find step-by-step tutorials for various HCI/BCI projects on our Instructables.

Below are some project ideas that you can try making at your home.

1. Controlling video game using brainwaves (EEG)
2. Visualising electrical impulses from eyes (EOG)
3. Recording EEG from visual cortex part of brain
4. Recording EEG from prefrontal cortex part of brain
5. Eye blink detection
6. Creating a drowsiness detector
7. Record publication-grade ECG
8. Measuring heart rate
9. Detecting heart beats
10. Record publication-grade EMG
11. Detecting up and down movement of eyes

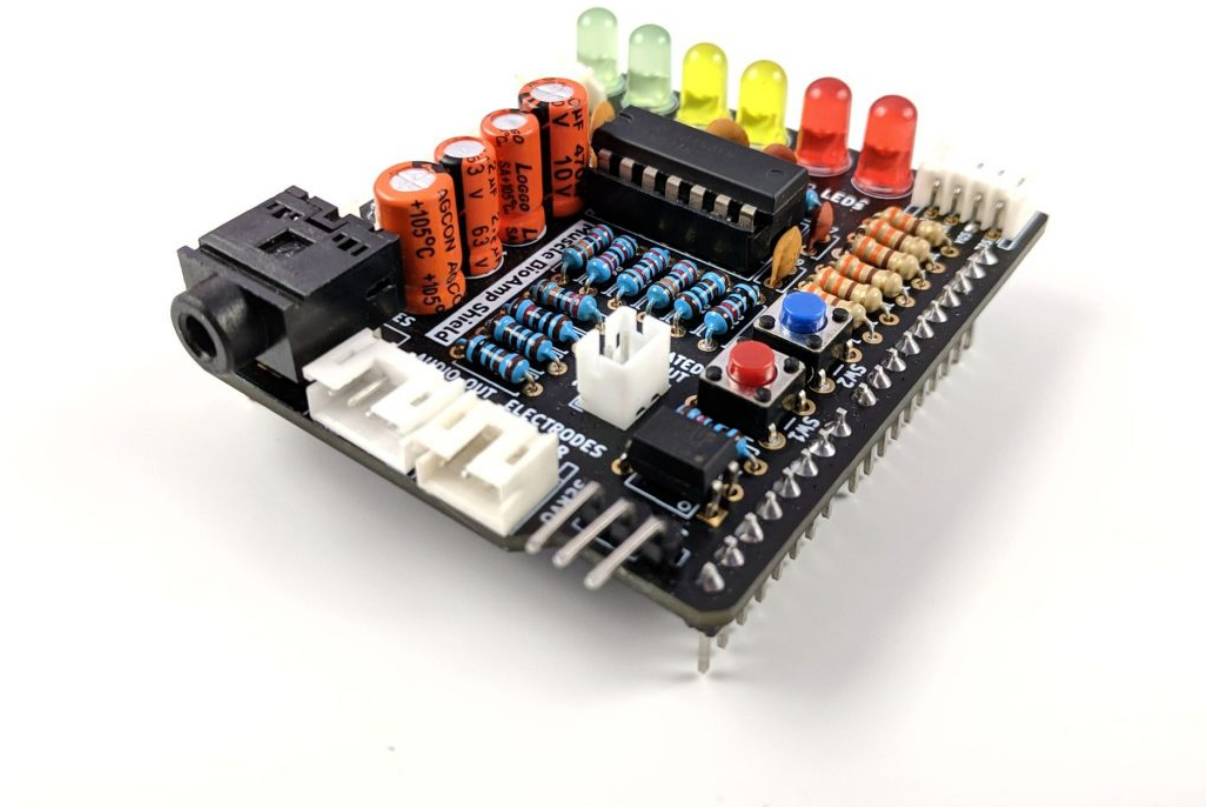
These are some of the project ideas but the possibilities are endless. So create your own Human Computer Interface (HCI) and Brain Computer Interface (BCI) projects and share them with us at contact@upsidedownlabs.tech

# 3 Muscle BioAmp Shield

v0.3

## 3.1 Overview

Muscle BioAmp Shield is an all-in-one Arduino Uno ElectroMyography (EMG) shield for learning neuroscience with ease which is inspired from BackYard Brains (BYB) Muscle Spiker shield and provides similar features like hobby servo output, user buttons, LED Bar, Audio output, and battery input. It is perfect for beginners as they can easily stack it on top of Arduino Uno to record, visualize and listen to their muscle signals to make amazing projects in the domain of Human-Computer Interface (HCI).




## 3.2 Features & Specifications

Muscle BioAmp Shield comes with various plug-and-play options so you can connect hundreds of extension boards like OLED screens, character displays, accelerometers, and servo controllers to name just a few using the STEMMA I2C interface. You also get STEMMA digital and STEMMA analog ports. On STEMMA analog port you can connect additional BioAmp EXG Pill or any other sensor with analog output. On STEMMA digital port you can connect any digital sensor or actuator of your choice.

| | |
|---|---|
| Input Voltage | 5V |
| Input Impedance | 10^11 ohm |
| Fixed Gain | x2420 |
| Bandpass filter | 72 – 720 Hz |
| Compatible Hardware | Arduino UNO |
| BioPotentials | EMG (Electromyography) |
| No. of channels | 1 |
| Electrodes | 3 (Positive, Negative, and Reference) |
| Dimensions | 6.0 x 5.3 cm |
| Open Source | Hardware + Software |

## 3.3 Hardware

Images below shows a quick overview of the hardware design.

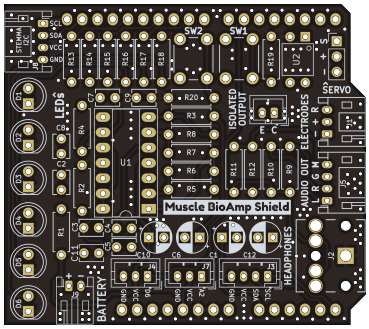



Fig. 1: *PCB Front*

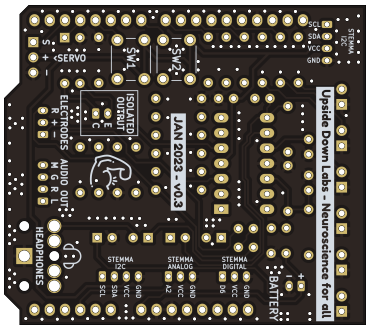



Fig. 2: *PCB Back*

## 3.4 Contents of the kit

There are 2 variants available for Muscle BioAmp Shield v0.3 kit - one comes with the shield assembled and the other one contains bare PCB of the sensor and the components separately which you can assemble pretty easily.

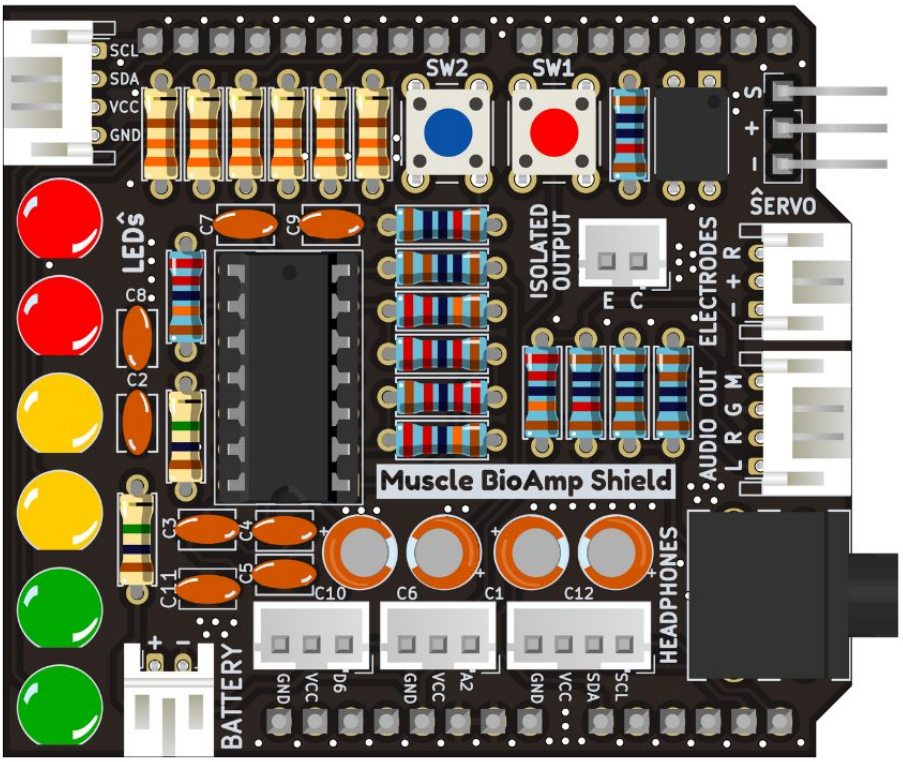



Fig. 3: *Assembled PCB*

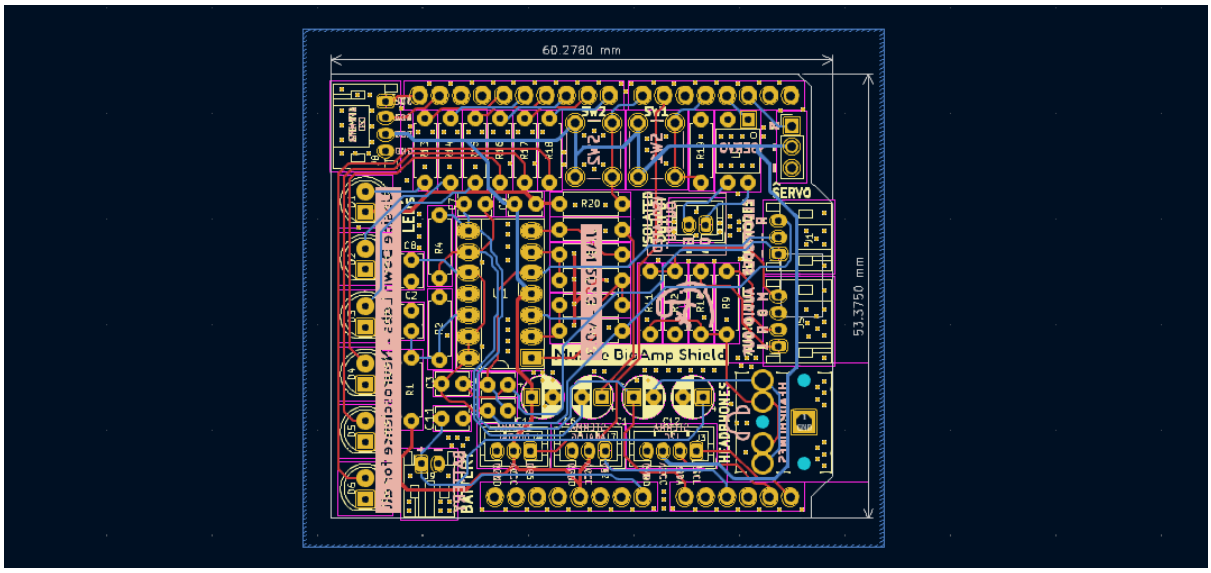



Fig. 4: *PCB Layout*

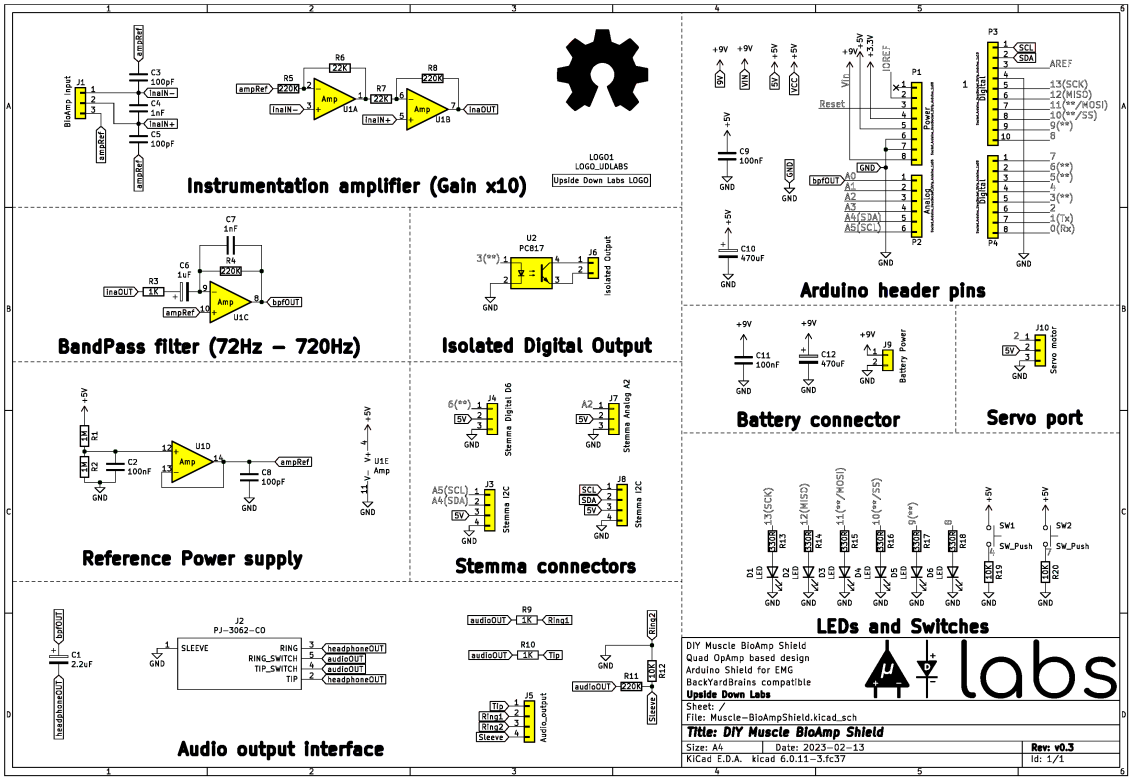



Fig. 5: *Schematic Diagram*

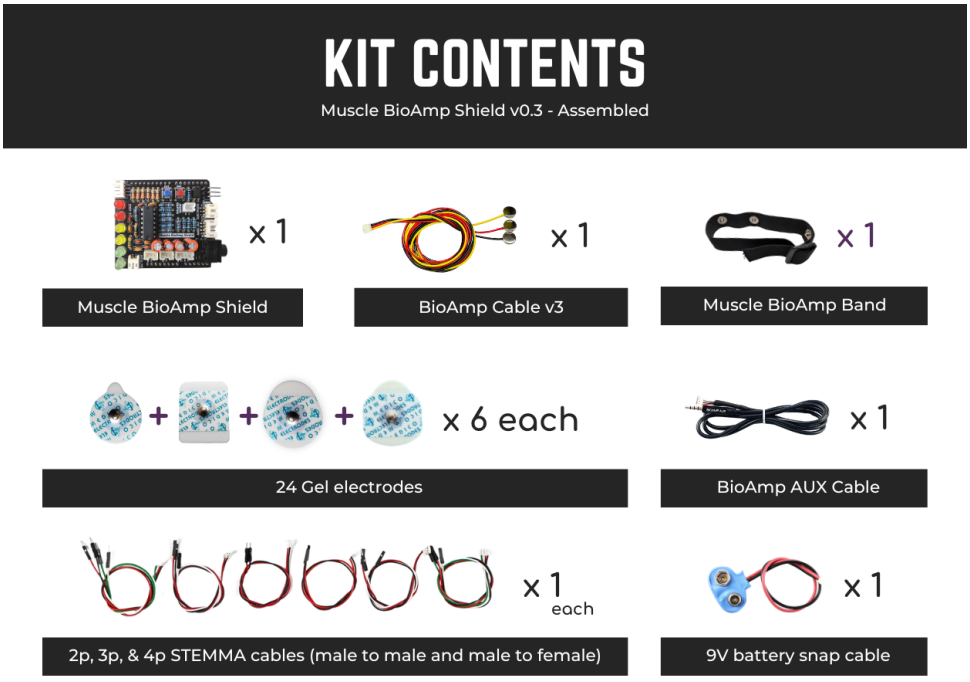
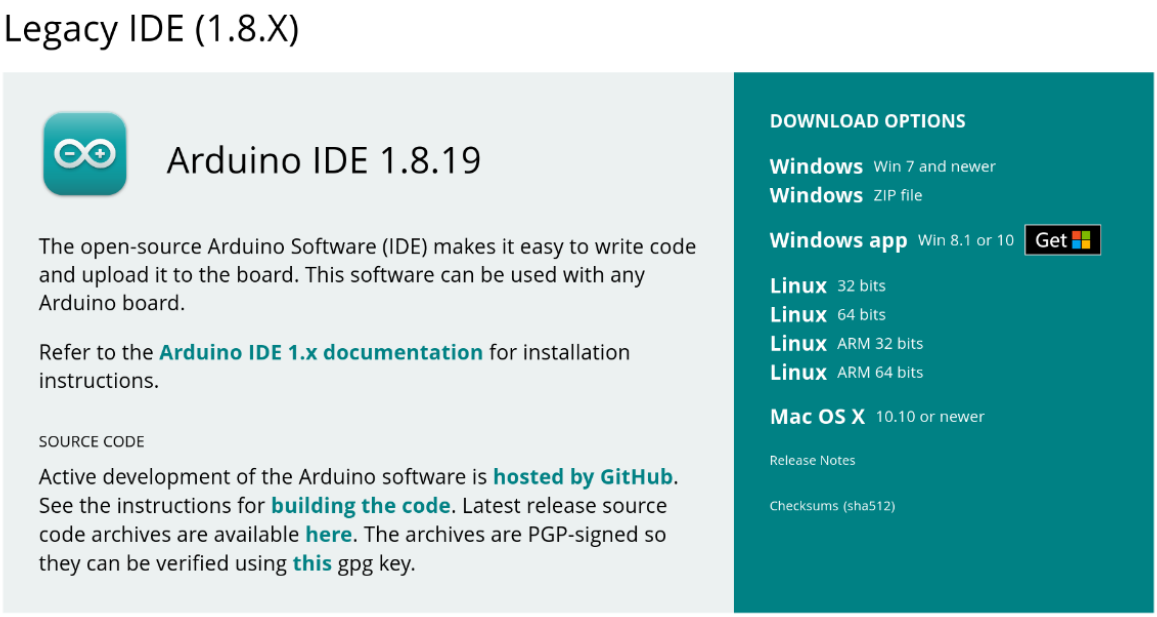



Fig. 6: *Assembled Muscle BioAmp Shield kit content*

## 3.5 Software requirements

1. **Arduino IDE**
   - Latest Version (v2.3.8+): Arduino IDE Latest Version Download *(Recommended for faster compilation and enhanced support for the latest Arduino UNO Development Boards)*
   - Legacy Version (v1.8.19): Arduino IDE v1.8.19 (legacy IDE) Download *(For Real-time signal visualization using a better built-in Serial Plotter)*




2. **Chords Web Visualizer**

   Use Chords Web, our custom web interface designed to visualize biopotential signals data directly in your browser. *(Plug-and-play signal visualization directly in any modern web browser.)* Flash the Chords-Arduino-Firmware on your board to visualize signals on Chords Web.

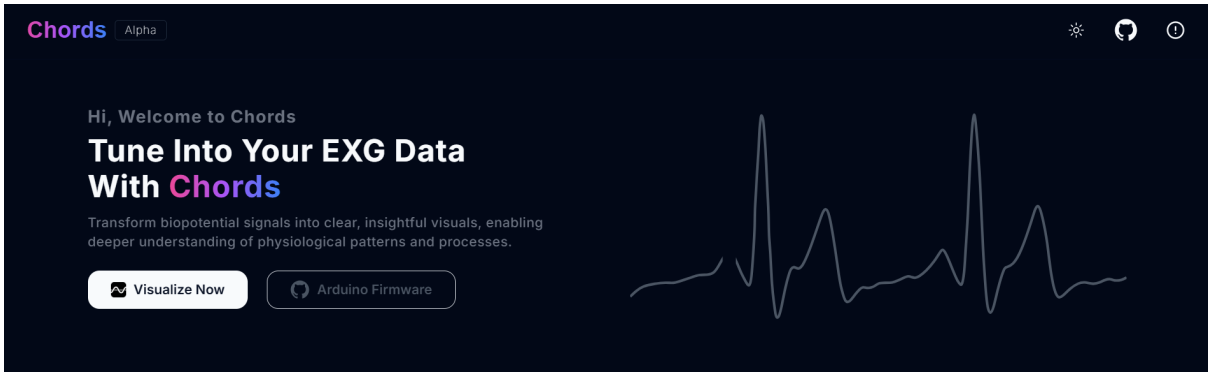


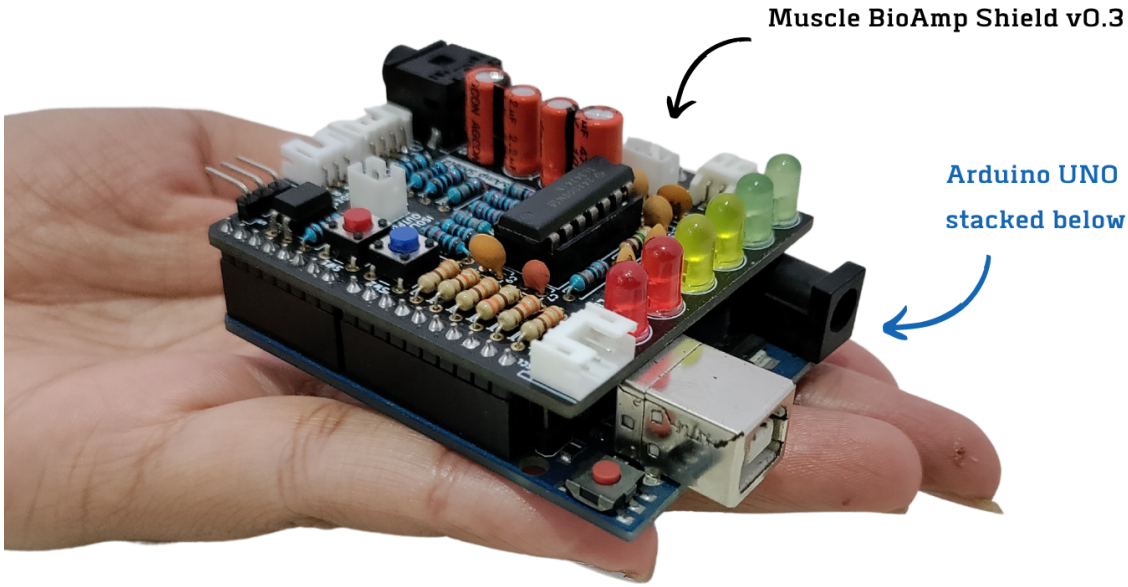

Fig. 7: Chords Web Landing Page

> **Note**
>
> For detailed instructions and visualization tips, check the Chords Web Documentation.
>
> https://youtu.be/IVIPnk9z75g

## 3.6 Using the Sensor

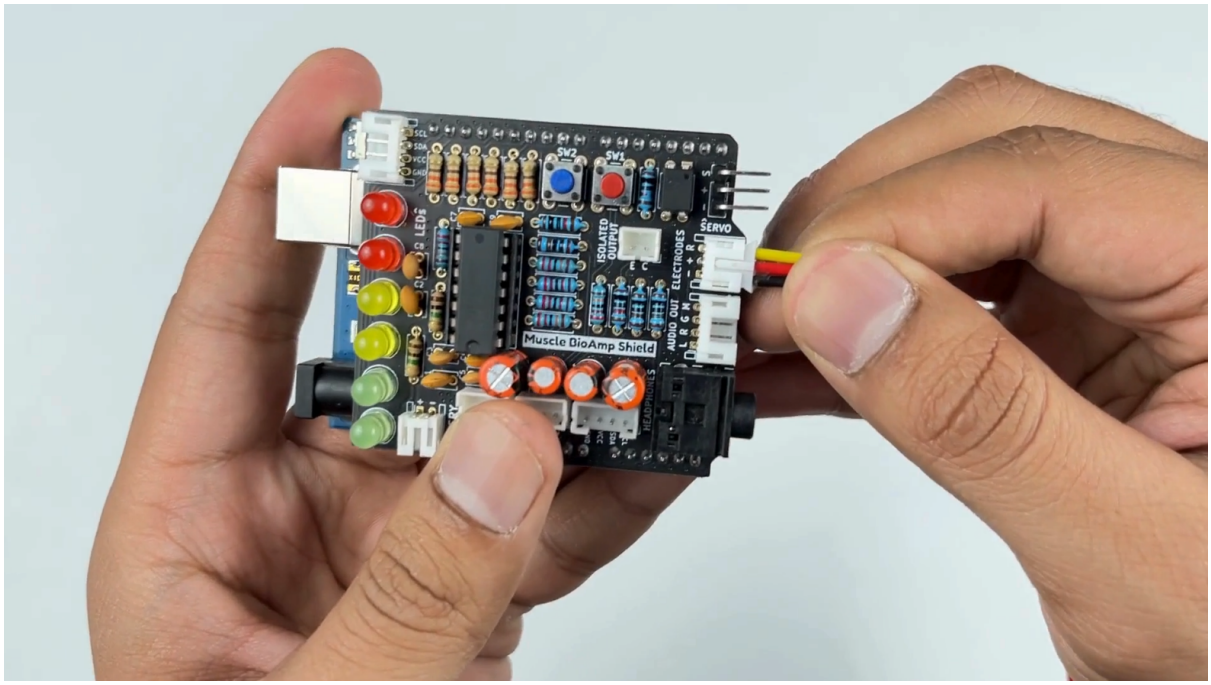
### 3.6.1 Step 1: Stack on Arduino Uno

Stack the Muscle BioAmp Shield on top of Arduino Uno properly.




### 3.6.2 Step 2: Connecting Electrode Cable

Connect the BioAmp Cable to Muscle BioAmp Shield as shown.



### 3.6.3 Step 3: Skin Preparation

Apply Nuprep Skin Preparation Gel on the skin surface where electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step Skin Preparation Guide.

### 3.6.4 Step 4: Electrode Placements

We have 2 options to measure the EMG signals, either using the gel electrodes or using dry electrode based Muscle BioAmp Band. You can try both of them one by one.

#### Using gel electrodes

1. Connect the BioAmp cable to gel electrodes,
2. Peel the plastic backing from electrodes
3. Place the IN+ and IN- cables on the arm near the ulnar nerve & REF (reference) at the back of your hand as shown in the connection diagram.

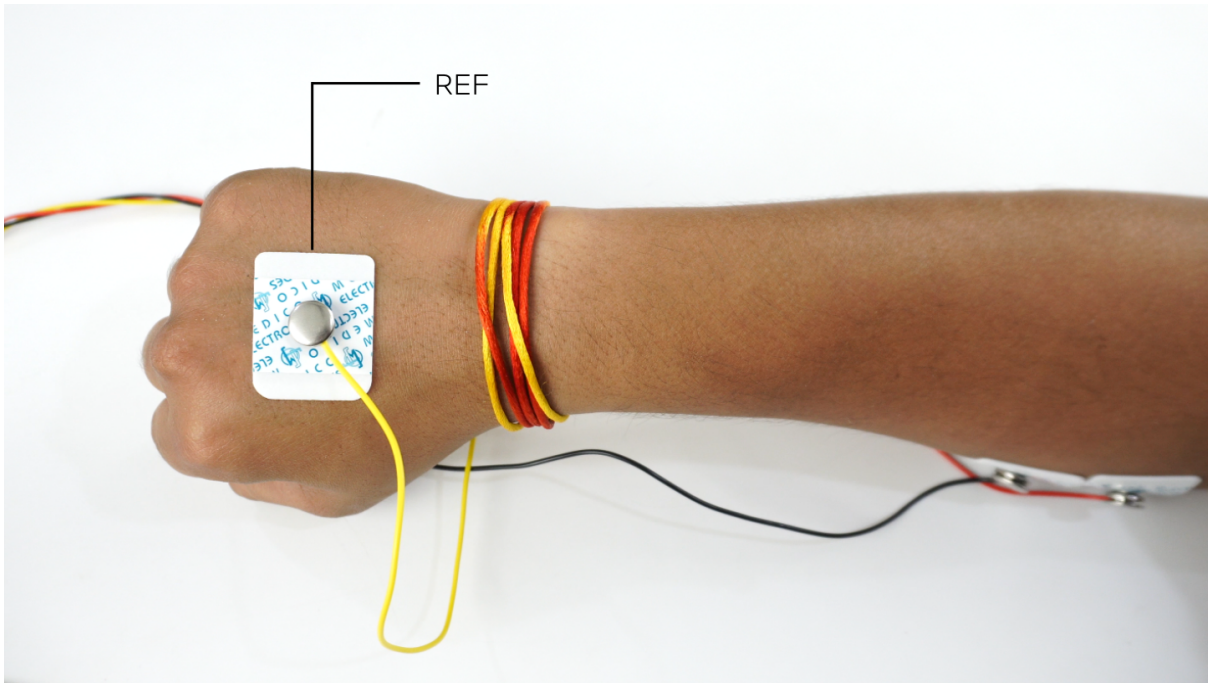



Fig. 8: *Electrode placement for REF cable*

#### Using Muscle BioAmp Band

1. Connect the BioAmp cable to Muscle BioAmp Band in a way such that IN+ and IN- are placed on the arm near the ulnar nerve & REF (reference) on the far side of the band.
2. Now put a small drop of electrode gel between the skin and metallic part of BioAmp cable to get the best results.

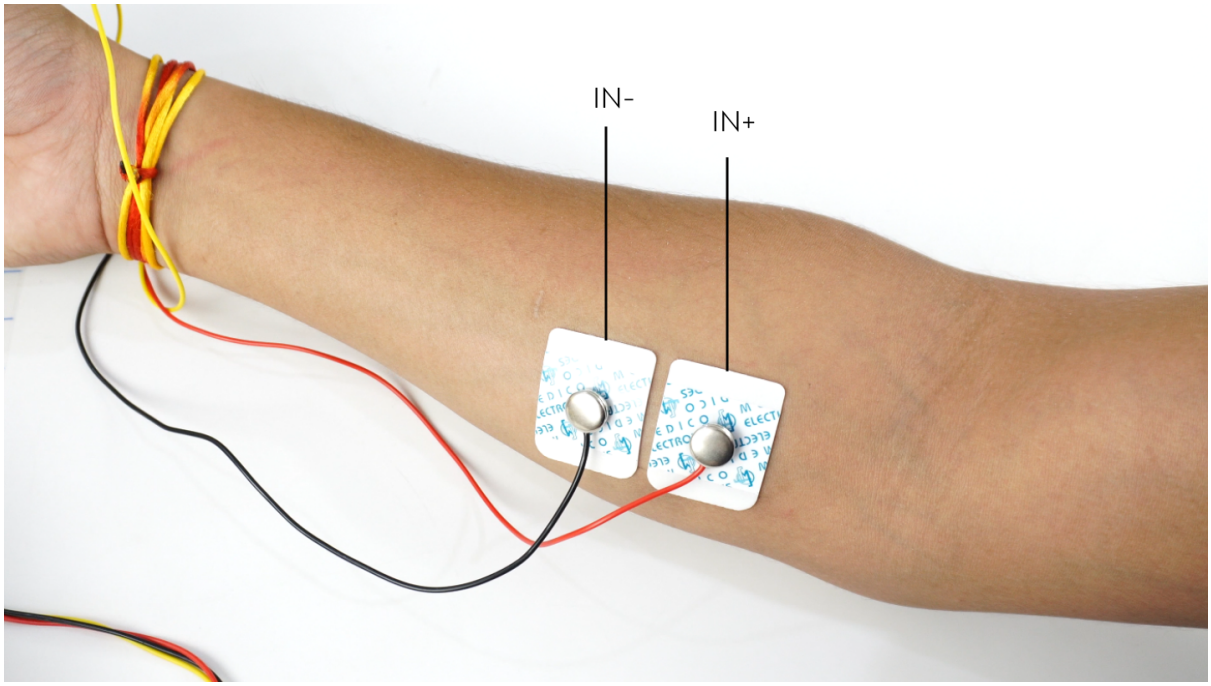



Fig. 9: *Electrode placement for IN+, IN- cables*

**Tip**

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

**Tutorial on how to use the band:**

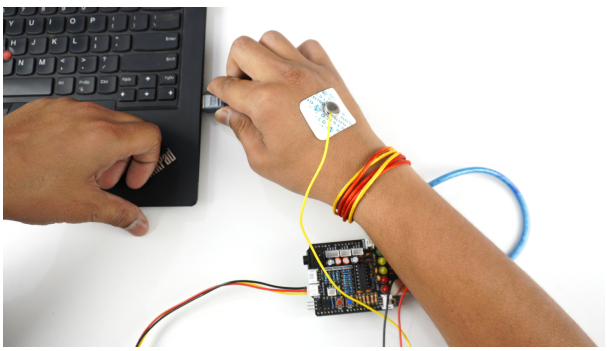
https://youtu.be/xYZdw0aesa0

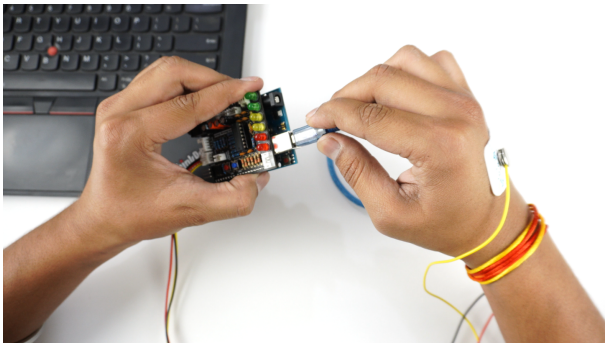
**Note**

In this demonstration we are recording EMG signals from the ulnar nerve, but you can record EMG from other areas as well (biceps, triceps, legs, jaw etc) as per your project requirements. Just make sure to place the IN+, IN- electrodes on the targeted muscle and REF on a bony part.

### 3.6.5 Step 5: Uploding the Code

- Connect your Arduino UNO to your laptop using the USB cable.





- Choose and copy one of the following Arduino sketches based on your requirements:

EMG Filter

EMG Envelope

To upload the code use Arduino IDE latest version (v2.3.8+):

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).
- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

**Warning**

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

### 3.6.6 Step 6: Visualise EMG signals on laptop

1. **Using Chords Web**

   For visualizing the EMG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

2. **Using Arduino legacy v1.8.19**

   To visualize the signal on Arduino v1.8.19 we will use inbuilt Serial Plotter:

- Navigate to (`Tools > Board`) and select your Development board or Arduino UNO Board (if not detected automatically).
- Open the Serial Plotter (`Tools > Serial Plotter` or `Ctrl+Shift+L`) to visualize the EMG signals.
- Once the Serial Plotter is open, ensure the baud rate is set to `115200`.
- Now flex your arm to visualize the muscle signals in real time on your laptop.

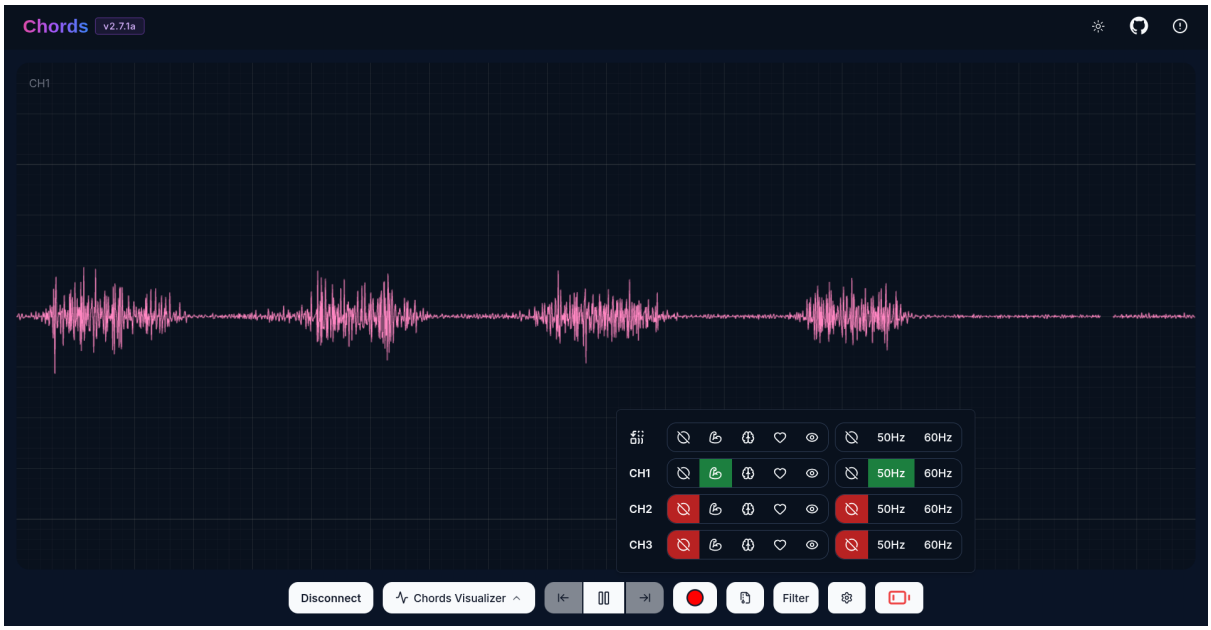



Fig. 10: Visualizing EMG signals on Chords Web

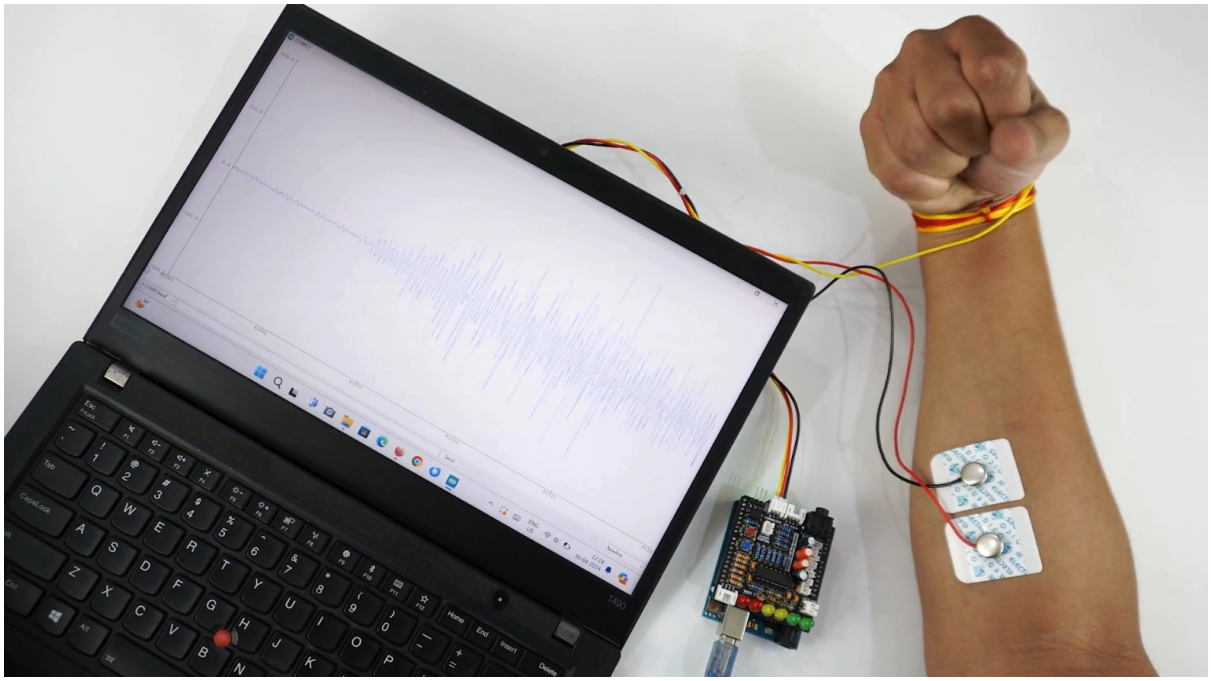


Fig. 11: *Visualise EMG signals on laptop*

### 3.6.7 Step 7: Visualise EMG signals on LEDs

Copy paste the Arduino Sketch given below in Arduino IDE:

LED Bar Graph

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm. You'll see the LED bar going up. More strength you apply, more the LED bar goes up.

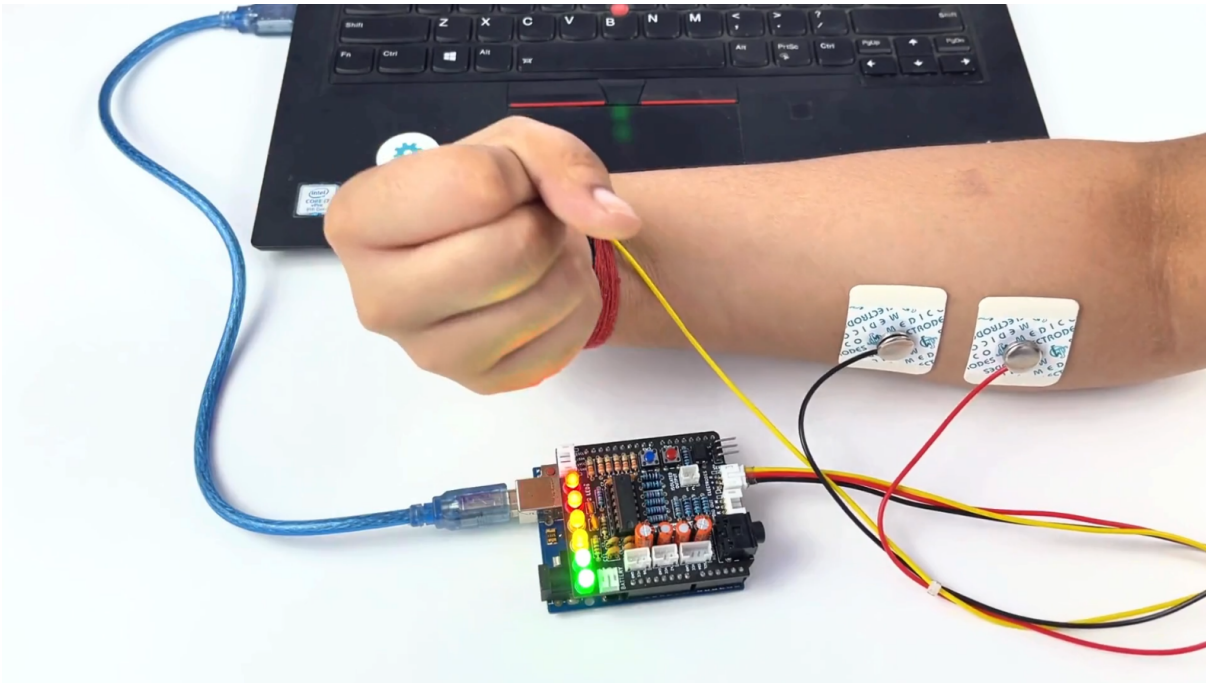


Fig. 12: *Visualise EMG signals on LEDs*

### 3.6.8 Step 8: Listen to your EMG signals

You can either listen to the muscle signals (EMG) on a speaker or wired earphones/headphones. Let's try both of them.

#### Listening EMG on speakers

1. Connect the BioAmp AUX cable on a bluetooth speaker that have 3.5mm jack support.
2. Switch on the speaker and turn the volume to maximum.
3. Flex and listen to your muscles.

#### Listening EMG on a wired earphones/headphones

1. Plug your wired earphones or headphones on the 3.5mm jack of BioAmp v1.5.
2. Plug it in your ears.
3. Flex and listen to your muscles.

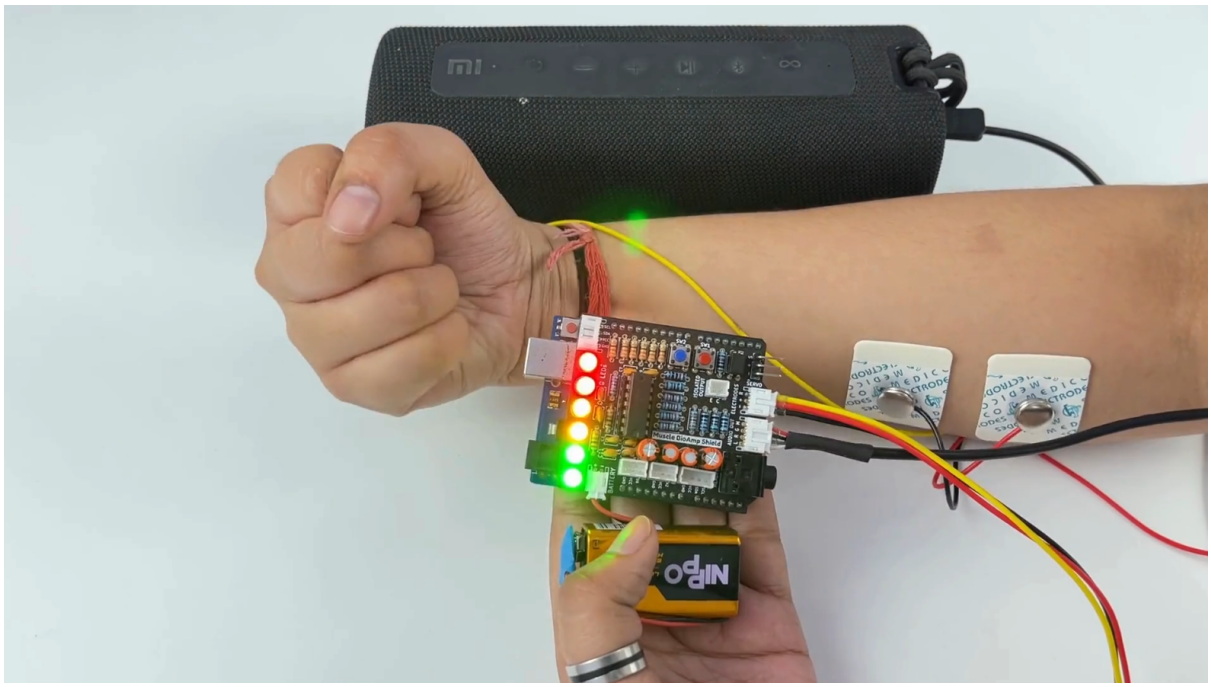


Fig. 13: Listening EMG on speakers

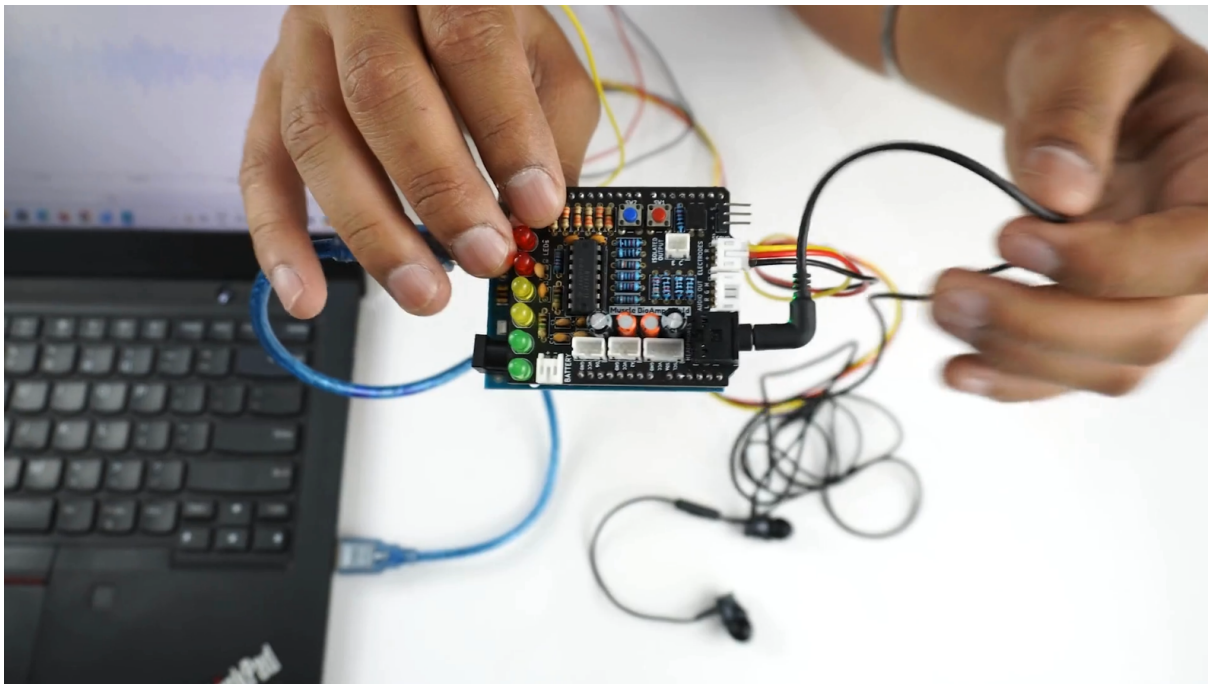


Fig. 14: *Listening EMG on a wired earphones/headphones*

### 3.6.9 Step 9: Controlling a servo motor

Connect the servo claw to Muscle BioAmp Shield.

Copy paste the Arduino Sketch given below in Arduino IDE:

Servo Controller

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm to control the servo claw in real time.

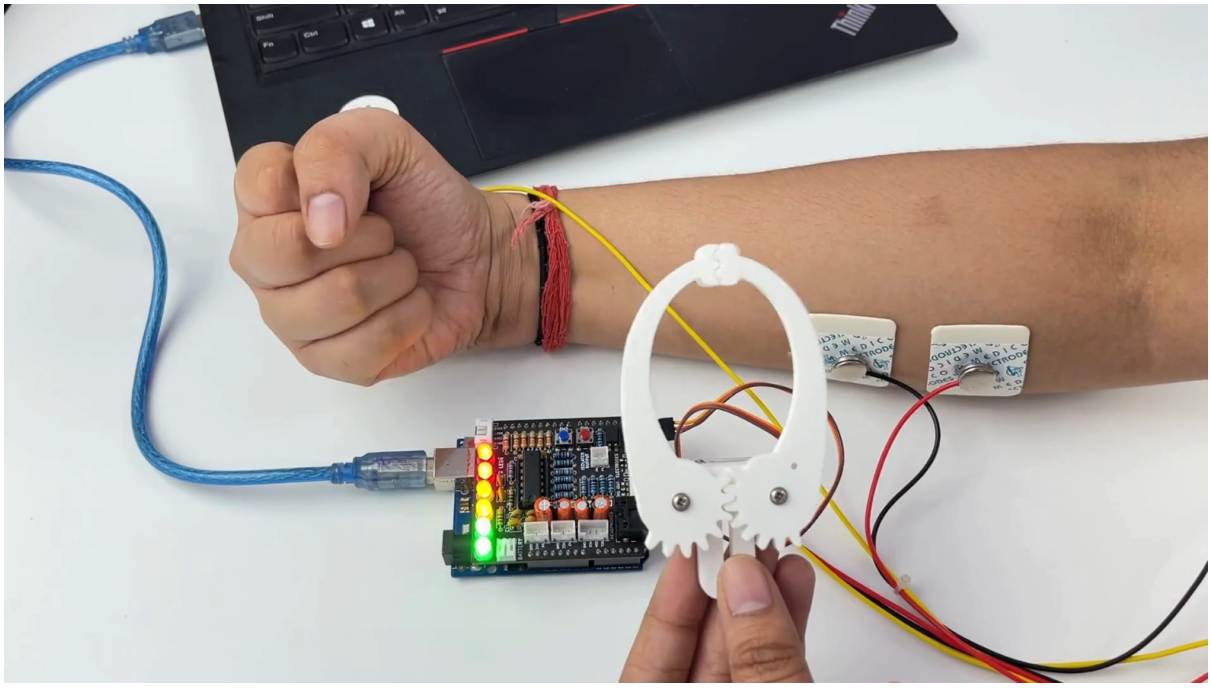
### 3.6.10 Step 10: Controlling a servo claw

Connect the servo claw to Muscle BioAmp Shield.

Copy paste the Arduino Sketch given below in Arduino IDE:

Claw Controller

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm to control the servo claw in real time.



### 3.6.11 Step 11: Connecting 9V battery

Till now, the power for the EMG system was coming from the laptop via USB cable of Arduino Uno but there can be 2 ways in which you can make the system portable:

- **Using 9V battery:** Directly connect a 9V battery to Muscle BioAmp Shield using a 9V snap cable.
- **Using Power Bank:** Instead of connecting the USB cable of Arduino Uno to laptop, you can directly connect it to power bank.

**Note**

Do not use 9V battery while controlling a servo claw using Muscle BioAmp Shield. Instead connect the Arduino UNO to a power bank or directly to your laptop.

### 3.6.12 Step 12: Other functionalities you can explore

#### Using I2C ports

There are 2 I2C ports available on Muscle BioAmp Shield and you can connect hundreds of devices having I2C compatibility using the 4-pin JST PH 2.0 mm STEMMA cables provided.

Some of the examples are: `OLED screens`, `character displays`, `temperature sensors`, `accelerometers`, `gyroscopes`, `light sensors`, `BioAmp Hardware`, etc.

#### Using STEMMA Digital port

Connect Arduino Uno's D6 digital I/O pins using STEMMA digital connectors.

#### Using STEMMA Analog port

Connect Arduino Uno's A2 analog input pins using STEMMA analog connectors.

#### Using user buttons

Program the 2 user buttons according to your project requirements.